

Hidden Region Finder

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ABSTRACT: The Method of Regions is a systematic way to derive asymptotic expansions of Feynman integrals. For Euclidean integrals there is a well-established algorithm that determines the complete set of regions from the facets of a Newton polytope defined by the graph polynomials. In many physically relevant Minkowski limits this construction misses additional contributions, known as hidden regions. A hidden region arises when an asymptotic scaling makes a set of monomials parametrically superleading, but these enhanced monomials cancel collectively at a pinch singularity on the contour of positive Schwinger parameters. Their sum is thereby reduced to the same order as terms which appeared subleading before the cancellation.

We present the Hidden Region Finder, a parameter-space algorithm that identifies such pinch singularities and determines the associated hidden regions and their scaling laws.

We explore massless four-, five- and six-point integrals in a variety of wide-angle, collinear, double-collinear and Regge limits. Across these different expansions, hidden regions recur in the same small set of seed topologies. This exploratory study showcases the potential of the Hidden Region Finder both to uncover missing contributions to asymptotic expansions and to reveal topology-based relations between physically distinct kinematic limits.

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1 Introduction

Hidden regions are contributions to asymptotic expansions that are invisible to the ordinary Newton-polytope formulation of the Method of Regions because nominally leading monomials cancel on a positive Landau pinch. We introduce the Hidden Region Finder (HRF), a parameter-space construction that determines the cancellation locus, the superleading and obstruction polynomials, the region scaling and the resolved leading Lee–Pomeransky polynomial. It handles general polynomial cancellation ideals, contraction-boundary regions, asymptotic-order alignment and successive cancellation layers. We develop the construction for massless internal propagators, where the Symanzik polynomials are linear in every individual Schwinger parameter.

Applications reveal a sparse topology-based organization that is not apparent from individual region searches. In the audited on-shell four-point sample through four loops, every certified wide-angle or three-channel Regge hidden region is associated with the three-loop Crown, either in the graph interior or as a contraction minor; an exhaustive analysis of 16 mixed-sign four-loop topologies without a Crown minor finds none. These results support both the Crown-minor and wide-angle-to-Regge conjectures in this sample. In addition, the five-loop SuperCrown supplies a positive contraction-minor test beyond the four-loop audit. In five-point scattering, the same two-loop seed supports distinct HRs in a spacelike-collinear expansion and in a composite multi-Regge limit with a soft central emission. Complete edge-flow tables and a light-cone pole-pinch audit determine the local Glauber modes and make the momentum-space reconstruction mode complete; a vertex-corrected spacelike descendant also shows why genuinely non-binomial cancellation factors are required. At six points, asymptotic-order alignment exposes a common positive cancellation hypersurface for NMRK with $w = z$ and for the double spacelike-collinear limit. The two expansions share the same factorized superleading support but resolve it at different cancellation depths, and therefore have distinct region vectors. Thus HRF is not only a region-finding algorithm: it exposes seed topologies and relations between kinematic expansions. The Method of Regions (MoR) constructs the asymptotic expansion of a multiscale Feynman integral as a sum of integrals associated with distinct scalings of the integration variables. Its geometric formulation in parameter space maps every monomial of the Lee–Pomeransky polynomial to an exponent point. Lower facets of the resulting Newton polytope determine the ordinary endpoint regions and their scaling vectors.

The Newton polytope retains the monomial exponents but not the information carried by their coefficients. In a physical Minkowski region, monomials of opposite sign can cancel at positive values of the parameters. If the cancellation is reinforced by a Landau pinch, a set of monomials may be individually more singular than the physical leading contribution while their sum is suppressed. The corresponding scaling is not a lower-facet normal of the

original polynomial and is therefore hidden from the ordinary geometric construction. We call it a *hidden region* (HR).

A hidden region is specified by more than a scaling vector. One must identify the common positive cancellation locus, determine which monomials form the superleading sector, resolve their cancellation against the remaining Lee–Pomeransky polynomial, and verify that the resulting leading integral is scaleful. The Hidden Region Finder (HRF) is the systematic parameter-space construction that performs these tasks.

The present paper concerns massless internal propagators only. This is a structural restriction, not a choice of examples: both Symanzik polynomials are then multiaffine, or equivalently linear in every individual edge parameter. Internal masses add $\mathcal{U} \sum_e m_e^2 x_e$ to $\mathcal{F}$, which can generate quadratic dependence on an individual x_e . Hidden regions in that more general setting are important, but require a separate extension of the construction and will not be treated here.

The presentation has two parts. Part I defines the construction without reference to a particular graph. Part II develops the same ideas through a sequence of massless examples. Each example introduces one additional phenomenon: a factorised cancellation with uniform scaling; an obstruction that fixes a non-uniform scaling; non-binomial cancellation factors; and finally composite limits for which preliminary asymptotic-order alignment and, in some cases, more than one cancellation layer are required. The examples also reveal a recurring physical organisation: a special seed topology can support different HRs in different kinematic expansions.

2 Hidden Regions Finder: Principles

2.1 Parameter-space setup and definitions

Consider a graph $\mathcal{G}$ with N massless internal propagators and Lee–Pomeransky (LP) polynomial

$$\mathcal{P}(\mathbf{x}, \mathbf{s}) = \mathcal{U}(\mathbf{x}) + \mathcal{F}(\mathbf{x}; \mathbf{s}). \quad (2.1)$$

The spanning-tree and spanning-2-forest formulae make the relevant structure immediate. For an L -loop graph, $\mathcal{U}$ is homogeneous of degree L , while $\mathcal{F}$ is homogeneous of degree $L + 1$ in the edge parameters. Each edge occurs at most once in every complementary tree or 2-forest product. Hence, for massless internal propagators, both graph polynomials are linear in each individual x_e . This multiaffinity is a standing assumption below. Up to a normalisation which is irrelevant here, its integral has the form

$$I_{\mathcal{G}} = \mathcal{N}_{\text{LP}} \int_0^\infty \prod_{e=1}^N \frac{dx_e}{x_e} x_e^{\nu_e} \mathcal{P}(\mathbf{x}, \mathbf{s})^{-D/2}. \quad (2.2)$$

The N variables x_e are independent Lee–Pomeransky edge parameters, each integrated over $(0, \infty)$. In particular, there is no constraint on $\sum_e x_e$, or on the sum of any subset of the parameters. We use x_e rather than the conventional Feynman-parameter symbol α_e .

An optional, distinct projective representation can be obtained by introducing one additional parameter x_0 and the homogeneous polynomial

$$\widehat{\mathcal{P}}(x_0, \mathbf{x}; \mathbf{s}) = \mathcal{F}(\mathbf{x}; \mathbf{s}) + x_0 \mathcal{U}(\mathbf{x}). \quad (2.3)$$

There are then $N + 1$ parameters, and their common rescaling may be fixed by inserting $\delta(1 - E(x_0, \mathbf{x}))$ for any positive degree-one homogeneous function E . This homogenised embedding is distinct from the non-projective LP integration space in (2.2) and is not assumed here.

Let $\delta \rightarrow 0^+$ define the kinematic expansion, and let $\mathbf{s}$ denote a set of real kinematic coordinates. No sign convention is imposed on the individual components of $\mathbf{s}$; a physical domain $\mathcal{K}$ is specified by the appropriate inequalities among them. This allows the same variables to be retained across different physical regions, making changes of sign under crossing or analytic continuation explicit. A separate positive parametrisation of a particular domain may be introduced locally as a bookkeeping aid for proving inequalities or performing semialgebraic sign tests, but neither the Newton-polytope construction nor the first-sheet pinch condition requires it. Retaining signed common variables also makes a change in the existence of a cancellation directly visible in a factorised superleading polynomial. After expanding the kinematics, write

$$\mathcal{P}(\mathbf{x}, \delta; \mathbf{s}) = \sum_i m_i, \quad m_i \equiv c_i(\mathbf{s}) \delta^{a_i} \mathbf{x}^{\mathbf{r}_i}, \quad \mathbf{x}^{\mathbf{r}_i} = \prod_{e=1}^N x_e^{r_{i,e}}. \quad (2.4)$$

Thus $\mathbf{r}_i = (r_{i,1}, \dots, r_{i,N})$ is the length- N LP-parameter exponent vector of the monomial m_i , while its augmented length- $N + 1$ exponent point is

$$\vec{r}_i = (\mathbf{r}_i; a_i) \in \mathbb{R}^{N+1}. \quad (2.5)$$

No fixed sign of m_i on the whole physical domain is assumed. Although $\delta^{a_i} \mathbf{x}^{\mathbf{r}_i} > 0$ for $\delta > 0$ and $\mathbf{x} > 0$, the coefficient $c_i(\mathbf{s})$ may vanish and change sign within or between the domains under comparison. The same qualification applies if several literal monomials have been grouped into a kinematic coefficient or polynomial term. A region vector is represented by an augmented normal whose last component is normalised to one,

$$\vec{v}_R = (\mathbf{v}_R; 1), \quad x_e \mapsto \delta^{v_{R,e}} x_e, \quad (2.6)$$

and the weight of a monomial is

$$w_R(m_i) = \vec{v}_R \cdot \vec{r}_i = a_i + \mathbf{v}_R \cdot \mathbf{r}_i. \quad (2.7)$$

The last component is always displayed when a vector is meant to be a normal in augmented exponent space. A list containing only the first N entries is an LP-parameter scaling and not a fully normalised augmented normal.

For a cancellation-free polynomial, the monomials of minimum weight form a lower face of its Newton polytope. A codimension-one lower face has an inward-pointing normal whose last component is positive; after normalising that component to one, the normal gives the corresponding facet-region scaling.

2.1.1 Positive pinches

The LP variables are integrated on the positive contour $\mathcal{C}_{\text{LP}} = \mathbb{R}_{>0}^N$. An interior first-sheet pinch of a polynomial $\mathcal{Q}(\mathbf{x}; \mathbf{s})$ is therefore a solution

$$\mathcal{Q} = 0, \quad \frac{\partial \mathcal{Q}}{\partial x_e} = 0 \quad (e = 1, \dots, N), \quad \mathbf{x} \in \mathbb{R}_{>0}^N, \quad \mathbf{s} \in \mathcal{K}. \quad (2.8)$$

For a homogeneous polynomial, the derivative equations imply $\mathcal{Q} = 0$ by Euler's identity. On a contraction stratum the same equations are imposed only on the active, nonzero parameters.

A cancellation in the positive orthant requires terms of both signs. Consequently, every active derivative at an interior pinch must either vanish identically or contain terms of both signs. This elementary necessary condition is useful as a rapid prefilter, but it neither identifies the common pinch locus nor proves that a hidden region exists. The defining conditions below first use the scaling law to isolate $\mathcal{F}_{\text{SL}}$, and then require a simultaneous positive solution of its derivative equations.

2.1.2 Definition of a hidden region

A HR is characterised by a superleading monomial support $\mathcal{S}_{\text{SL}}$, its associated polynomial

$$\mathcal{F}_{\text{SL}} = \sum_{i \in \mathcal{S}_{\text{SL}}} m_i, \quad (2.9)$$

and a final scaling

$$\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1) \quad (2.10)$$

such that:

- (a) every monomial indexed by $i \in \mathcal{S}_{\text{SL}}$ has the same individual weight,

$$w_{\text{HR}}(m_i) = W_{\text{SL}}, \quad i \in \mathcal{S}_{\text{SL}}; \quad (2.11)$$

- (b) $\mathcal{F}_{\text{SL}}$ has a common first-sheet pinch in $\mathcal{K}$,

$$\left. \frac{\partial \mathcal{F}_{\text{SL}}}{\partial x_e} \right|_{\mathbf{x} > 0, \mathbf{s} \in \mathcal{K}} = 0 \quad \text{for every active } x_e; \quad (2.12)$$

- (c) the cancellation suppresses the sum of the superleading monomials to the first nonvanishing LP layer of weight W_{HR} , with

$$W_{\text{HR}} > W_{\text{SL}}; \quad (2.13)$$

- (d) after resolving the cancellation, the leading LP polynomial defines a scaleful lower-facet region.

The gap $W_{\text{HR}} - W_{\text{SL}}$ measures the depth of the cancellation. It is a physical output of the full problem and is not a freely chosen normalisation.

The superleading sector is not defined merely by mixed signs. Mixed signs are necessary for an interior pinch, but the simultaneous positive solution, the hierarchy, and the final lower-facet condition are all essential.

Principle 1 (Scaling precedes the positive pinch test). *The mixed-sign test is applied to derivatives of the starting polynomial $\mathcal{F}_\star$, but the simultaneous Landau equations are not. The weight homogeneity and separation conditions must first isolate the support $\mathcal{S}_{\text{SL}}$. The positive first-sheet equations are then imposed on the resulting polynomial $\mathcal{F}_{\text{SL}}$. Generically $\partial\mathcal{F}_\star/\partial x_e = 0$ has no simultaneous positive solution because $\mathcal{F}_\star$ still contains the obstruction and other terms excluded by the HR scaling.*

2.2 The core HRF construction

The core construction receives a specified starting polynomial $\mathcal{F}_\star$ on a specified set of active LP parameters. It has two parts. The first constructs and validates a decomposition into a superleading cancellation sector and an obstruction. The second determines the scaling, resolves the complete LP polynomial and certifies the resulting lower facet. The choices of starting face and contraction stratum are outer operations discussed only after this core construction is complete. Part I may use homogeneous-weight and strict-separation equations to decide whether a proposed support defines a valid decomposition. This provisional selection is distinct from Part II, which determines and certifies the physical total region vector from the complete LP polynomial.

Before any LP-parameter scaling, expand

$$\mathcal{F}(\mathbf{x}, \delta; \mathbf{s}) = \mathcal{F}_0(\mathbf{x}; \mathbf{s}) + \sum_{n>0} \delta^n \mathcal{F}_n(\mathbf{x}; \mathbf{s}). \quad (2.14)$$

When the relevant superleading monomials are contained in $\mathcal{F}_0$, the cancellation analysis begins with $\mathcal{F}_\star = \mathcal{F}_0$.

2.2.1 Part I: constructing and validating the decomposition

The derivatives of $\mathcal{F}_\star$ are inspected for primitive, non-monomial polynomial factors $f_j(\mathbf{x}, \mathbf{s})$ which can vanish for $\mathbf{x} > 0$ and $\mathbf{s} \in \mathcal{K}$. At fixed $\mathbf{s} \in \mathcal{K}$, a necessary condition for such a zero is that the nonzero coefficients of the LP-parameter monomials are not all of the same sign. A monomial factor such as x_e is not a cancellation hypersurface in the interior and is removed from the primitive factor.

Literal factorisation and kinematic-coefficient harvesting are the first, inexpensive form of this derivative harvest. They need not expose every primitive normal polynomial: the derivatives may contain only invertible linear combinations of the normals. At this stage, however, the derivatives are inspected rather than imposed. In particular, one must not replace this harvest by the ideal generated by all derivatives of $\mathcal{F}_\star$. When an obstruction is present, its derivatives need not vanish on the true HR locus even though the derivatives of the subsequently selected $\mathcal{F}_{\text{SL}}$ do. Saturating that full gradient ideal would merely impose the wrong equations more efficiently. The harvest therefore retains candidate factors and kinematic sectors without requiring any complete derivative to vanish.

This inspection generates candidate cancellation structures; it is not a Landau test on $\mathcal{F}_\star$. In particular, the full derivative system of $\mathcal{F}_\star$ is not required to have a positive solution at this stage and generically does not. A generator-based candidate support is retained only if it admits a homogeneous-weight solution that strictly separates $\mathcal{S}_{\text{SL}}$ from

the obstruction and all other monomials. At this stage these linear conditions validate the candidate decomposition; they do not yet determine the certified physical total vector.

Generators and the obstruction decomposition. Candidate generators are products of genuine cancellation polynomials,

$$g_k = \prod_{f_j \in S_k} f_j, \quad (2.15)$$

with at least two cancellation factors in the generator structures used by HRF. The factors, not monomial edge parameters, define the singular hypersurfaces. For a generator set $G = \{g_1, \dots, g_r\}$, define

$$\mathcal{I}_{\text{canc}}(G) = \langle g_1, \dots, g_r \rangle. \quad (2.16)$$

The HRF seeks a decomposition

$$\mathcal{F}_\star = \mathcal{F}_{\text{SL}} + \mathcal{F}_{\text{Obs}}, \quad \mathcal{F}_{\text{SL}} \in \mathcal{I}_{\text{canc}}(G), \quad (2.17)$$

or equivalently

$$\mathcal{F}_{\text{SL}} = \sum_k M_k(\mathbf{x}, \mathbf{s}) g_k. \quad (2.18)$$

The polynomial $\mathcal{F}_{\text{Obs}}$ is the obstruction: it prevents the full starting polynomial from vanishing on the candidate cancellation locus but is to be suppressed relative to the individual SL monomials by the HR scaling.

Different generator presentations can describe the same ideal and the same pinch locus. The presentation is a discovery device and does not label a physical region. Polynomial multipliers M_k in (2.18) are allowed; a special factorised presentation valid for one topology must not be promoted to a general rejection rule.

Positive-pinch validation. Only for the selected polynomial $\mathcal{F}_{\text{SL}}$ is the simultaneous first-sheet pinch condition imposed. When its derivative equations reduce to a set of primitive cancellation factors, physical admissibility requires a common positive zero at one and the same kinematic point:

$$\exists (\mathbf{x}, \mathbf{s}) \in \mathbb{R}_{>0}^N \times \mathcal{K} : \quad f_1(\mathbf{x}, \mathbf{s}) = \dots = f_r(\mathbf{x}, \mathbf{s}) = 0. \quad (2.19)$$

Equivalently one solves (2.12) directly. The common locus, rather than any one factor in isolation, encodes the pinch. The elementary mixed-sign condition can discard impossible ingredients, but it cannot replace the scaling selection of $\mathcal{S}_{\text{SL}}$ or the subsequent simultaneous positive solution.

If literal factorisation is insufficient at this candidate-specific stage, one may complete the same calculation algebraically. Let us define the selected leading-sector gradient ideal $\mathcal{I}_{\nabla}^{\text{SL}}$ and the localization element S by

$$\mathcal{I}_{\nabla}^{\text{SL}} \equiv \left\langle \frac{\partial \mathcal{F}_{\text{SL}}}{\partial x_e} \right\rangle_e, \quad \text{and} \quad S \equiv \prod_{e \in \text{active}} x_e \prod_a h_a(\mathbf{s}), \quad (2.20)$$

where the $h_a(\mathbf{s})$ are kinematic factors known to be nonzero in the domain under consideration. Its interior, kinematically constrained form is

$$(\mathcal{I}_{\nabla}^{\text{SL}} + \mathcal{I}_{\text{kin}}) : S^{\infty}, \quad (2.21)$$

with $\mathcal{I}_{\text{kin}}$ imposing the algebraic constraints of the chosen expansion. Alternatively, the ideal may be generated by only the selected derivative-sector relations that the candidate requires. This saturation records that the factors in S cannot vanish in the interior and exposes consequences that otherwise require division by them. It is an optional completion or validation of an already selected cancellation sector, not a general pre-decomposition harvest of $\partial\mathcal{F}_{\star}$.

Dissection provides a complementary and especially useful validation when first encountering a new cancellation geometry. Starting from a common positive solution, choose coordinates u_{α} tangent to the pinch locus and independent normal coordinates $y_j = f_j$. Split a neighbourhood of the locus into sign sectors $y_j = \sigma_j t_j$, with $t_j \geq 0$ and $\sigma_j = \pm 1$, retaining the Jacobian of the change of variables and the portion for which every $x_e > 0$. In each such sector the cancellation locus is an endpoint and the transformed LP polynomial is treated by the ordinary geometric Method of Regions. Its lower Newton-polytope facets determine the local scalings; pulling their normals back to the original Schwinger parameters checks the candidate HR vector. Thus dissection tests the local cancellation geometry and the existence of a scaleful region without imposing the full gradient of an obstructed starting polynomial. The lower-facet certificate and its relation to a direct original-coordinate test are described in Sec. 2.2.2.

2.2.2 Part II: determining and certifying the scaling

The complete LP polynomial is decisive. For a candidate final vector, organise the complete polynomial into its actually occupied weight layers,

$$\mathcal{P}(\delta^v \mathbf{x}, \delta; \mathbf{s}) = \sum_{q \in \Lambda} \delta^q \mathcal{P}_q(\mathbf{x}, \mathbf{s}), \quad (2.22)$$

where Λ is the set of weights which occur. One must not insert fictitious intermediate layers. For example, if a parametrisation produces only even powers of the expansion parameter, the absence of an odd power is not an additional cancellation.

Let

$$\mathcal{I} = \langle f_1, \dots, f_r \rangle \quad (2.23)$$

be the ideal generated by independent cancellation factors defining the common pinch. A layer $\mathcal{P}_q$ which lies in $\mathcal{I}$ vanishes on the pinch; a layer in $\mathcal{I}^m$ vanishes to at least m -th order in transverse directions. The relevant information is its leading nonzero class in the filtration

$$\mathcal{I}^m / \mathcal{I}^{m+1}. \quad (2.24)$$

Locally, take transverse coordinates $y_j = f_j$. If the approach to the common pinch scales as

$$y_j \sim \delta^{\tau_j}, \quad \tau_j > 0, \quad (2.25)$$

then a jet $y_1^{\beta_1} \cdots y_r^{\beta_r}$ in the q -th occupied layer has effective weight

$$q + \sum_j \beta_j \tau_j. \quad (2.26)$$

The first resolved nonvanishing contribution, after accounting for these transverse weights and the $\mathcal{U}$ balance, defines W_{HR} . It need not be the first nominal obstruction layer. A second or later layer may also vanish on the same singular locus.

All independent transverse coordinates must be suppressed with positive weights. A solution which suppresses only a proper subset of the cancellation factors does not certify the common pinch under consideration. Likewise, an unresolved continuous family signals that the total scaling has not yet been fixed and may correspond to a residual scaleless rescaling.

Hierarchy conditions. The final, normalised vector $\vec{v}_{\text{HR}} = (v_{\text{HR}}; 1)$ must first satisfy

$$w_{\text{HR}}(m_i) = W_{\text{SL}}, \quad m_i \in \mathcal{F}_{\text{SL}}. \quad (2.27)$$

For the general, layered problem, the remaining hierarchy is a condition on the resolved jets rather than on the nominal weights of individual monomials. If $y^\beta h_\beta$ is a nonzero leading jet of an occupied layer $\mathcal{P}_q$, then

$$q + \beta \cdot \tau \geq W_{\text{HR}} > W_{\text{SL}}, \quad (2.28)$$

and equality must occur for at least one resolved contribution. A nominal layer with $q < W_{\text{HR}}$ is therefore allowed only if its positive ideal order supplies enough transverse suppression to satisfy (2.28). This is precisely how a second cancellation layer can occur.

If no intermediate ideal-valued layer is present, every term outside $\mathcal{F}_{\text{SL}}$ has zero transverse ideal order. In that special case (2.28) reduces to the single-layer monomial condition

$$w_{\text{HR}}(m_i) \geq W_{\text{HR}} > W_{\text{SL}}, \quad m_i \in \mathcal{P} \setminus \mathcal{F}_{\text{SL}}. \quad (2.29)$$

Equations (2.27)–(2.28), the positive transverse weights, and the balance with $\mathcal{U}$ determine the physical cancellation depth and remove any residual uniform rescaling ambiguity.

Terms suppressed at fixed LP parameters must be restored before this test. If a term from $\mathcal{F}_n$, $n > 0$, is promoted to the SL level or below by the candidate scaling, then the proposed $\mathcal{F}_{\text{SL}}$ is incomplete and the analysis must be repeated with that term included.

Scalefulness and dissection. Whether the region integral is scaleful is a property of the resolved monomial support of the LP polynomial. It does not depend on the space-time dimension D . A mere count of variables appearing in selected monomials is a useful diagnostic but is not a general certificate.

After the cancellations are resolved, the decisive geometric requirement is that the leading support define a lower facet with an inward-pointing normal whose last component is positive. There are two cases.

- (1) If the monomials at the resolved W_{HR} layer already exhibit the required lower facet in the original coordinates, this is a sufficient certificate and no change of variables is needed.
- (2) If that test is inconclusive, or if the full-layer equations do not fix a unique total vector, one must use local coordinates adapted to the pinch. A dissection separates the integration domain into sectors and maps the cancellation hypersurfaces to endpoints. In each sector the polynomial is cancellation-free, so the ordinary Newton-polytope facet criterion is both the safe region test and the direct route to the region integral.

The facet normal obtained after dissection is pulled back to the original LP coordinates and then normalised with last component one. This pullback is the authoritative total vector. It may differ from a provisional vector obtained from homogeneous weight equations if those equations retain a residual rescaling or have not resolved all cancellation layers.

Dissection is therefore not required for every discovery. A unique ideal-layer solution supplies the same certification in the original coordinates. Dissection remains the necessary fallback when the local ideal geometry is nonlinear, singular or non-unique, and it is often useful later for evaluating the HR integral.

2.3 Wrappers around the core construction

Two extensions change the problem supplied to the core without changing its internal logic. Asymptotic-order alignment selects a nontrivial starting face of the full δ -dependent polynomial. A boundary search first restricts the graph polynomial to a contraction stratum. Either operation, or both together, is followed by the same decomposition and scaling construction of Sec. 2.2.

2.3.1 Asymptotic-order alignment

In a composite limit, the relevant cancellation structure may not be visible in the naive fixed- $\mathbf{x}$ leading polynomial $\mathcal{F}_0$. A preliminary LP-parameter rescaling can expose a different face of the full δ -dependent polynomial before the cancellation decomposition is applied. This operation is called *asymptotic-order alignment*.

The rescaling changes the effective expansion order of a monomial from a_i to $a_i + \phi \cdot \mathbf{r}_i$. Monomials that belong to different fixed- $\mathbf{x}$ orders in δ can thereby acquire a common effective weight and enter the same cancellation sector. This cross-order alignment, rather than the mere selection of a different face, is the purpose of the operation.

Let $\phi \in \mathbb{Q}^N$ be the preliminary alignment vector and introduce new parameters $\mathbf{y}$ by

$$x_e = \delta^{\phi_e} y_e. \quad (2.30)$$

For a monomial $m_i = c_i(\mathbf{s}) \delta^{a_i} \mathbf{x}^{\mathbf{r}_i}$, define

$$\omega_\phi(m_i) = a_i + \phi \cdot \mathbf{r}_i. \quad (2.31)$$

The order-aligned starting polynomial is the exposed face

$$\mathcal{F}^{[\phi]} = \sum_{\substack{m_i \in \mathcal{F} \\ \omega_\phi(m_i) = \min_{m_j \in \mathcal{F}} \omega_\phi(m_j)}} m_i. \quad (2.32)$$

The general HRF starting object is therefore

$$\mathcal{F}_\star = \begin{cases} \mathcal{F}_0, & \text{ordinary search,} \\ \mathcal{F}^{[\phi]}, & \text{order-aligned search.} \end{cases} \quad (2.33)$$

The same cancellation-factor, positive-pinch and obstruction analysis is then applied to $\mathcal{F}_\star$.

Asymptotic-order alignment is not itself the hidden-region scaling. It is the first of two coordinate transformations. It exposes the polynomial on which monomials from different original δ -orders can participate in one cancellation; HRF then determines the scaling about that cancellation locus in the aligned coordinates.

The second vector and the composition law. Let $\boldsymbol{\rho} \in \mathbb{Q}^N$ denote the HRF scaling in the aligned variables,

$$y_e = \delta^{\rho_e} z_e. \quad (2.34)$$

Combining Eqs. (2.30) and (2.34) gives

$$x_e = \delta^{\phi_e + \rho_e} z_e, \quad \mathbf{v}_{\text{HR}} = \boldsymbol{\phi} + \boldsymbol{\rho}, \quad \vec{v}_{\text{HR}} = (\boldsymbol{\phi} + \boldsymbol{\rho}; 1). \quad (2.35)$$

Only the N edge-parameter exponents are added. The last component is the normalisation of the single expansion parameter and is appended once after the two transformations have been composed. Thus one must never add $(\boldsymbol{\phi}; 1)$ and $(\boldsymbol{\rho}; 1)$, which would spuriously produce a last component equal to two.

The composition can be checked without reference to any factorisation. The same original monomial has the successive weights

$$a_i + \boldsymbol{\phi} \cdot \mathbf{r}_i, \quad a_i + (\boldsymbol{\phi} + \boldsymbol{\rho}) \cdot \mathbf{r}_i. \quad (2.36)$$

The second expression, evaluated for every monomial of the complete LP polynomial, is the absolute weight relevant to the physical region. Stage-local weights obtained after an overall power of δ has been factored from $\mathcal{F}^{[\phi]}$ cannot be added to, or directly identified with, these absolute weights.

There is one important qualification. A run on the isolated aligned face may first return a provisional vector $\boldsymbol{\rho}^{(0)}$. This is the correct second vector only if restoration of the complete LP polynomial does not promote further occupied layers. In the general case HRF must restore all layers, compute their orders in the cancellation ideal, and solve the resolved hierarchy again. The resulting $\boldsymbol{\rho}$, rather than $\boldsymbol{\rho}^{(0)}$, is the vector to use in Eq. (2.35). Successive cancellation layers can change a non-uniform component of the second vector; their effect is not, in general, reducible to an overall shift.

Uniform-shift ambiguity. The second Symanzik polynomial is homogeneous of fixed parameter degree. Consequently,

$$\boldsymbol{\phi} \mapsto \boldsymbol{\phi} + c(1, \dots, 1) \quad (2.37)$$

adds the same amount to the weight of every monomial of $\mathcal{F}$ and does not change the face $\mathcal{F}^{[\phi]}$. The preliminary alignment vector is therefore an equivalence class, not a unique vector.

The coordinate transformation itself shows how this ambiguity is removed. For fixed original $\mathbf{x}$, replacing ϕ by $\phi + c\mathbf{1}$ replaces the aligned coordinates by $\mathbf{y}' = \delta^{-c}\mathbf{y}$. The same second transformation is therefore represented by

$$\boldsymbol{\rho}' = \boldsymbol{\rho} - c\mathbf{1}, \quad (\phi + c\mathbf{1}) + \boldsymbol{\rho}' = \phi + \boldsymbol{\rho}. \quad (2.38)$$

Hence the physical total vector is independent of the chosen alignment representative, provided the second vector is transformed consistently.

In practice this compensation is fixed by the complete LP polynomial. The polynomials $\mathcal{U}$ and $\mathcal{F}$ have degrees L and $L + 1$, respectively, so a uniform shift changes their relative weights. Requiring the resolved cancellation layer to balance the appropriate $\mathcal{U}$ layer, while satisfying the hierarchy conditions of Sec. 2.2.2, fixes the representative and the physical gap $W_{\text{HR}} - W_{\text{SL}}$. This gap is a property of the region and is not an adjustable normalisation.

The consistent two-stage procedure is therefore:

- (i) choose any convenient representative ϕ that exposes the required face;
- (ii) run HRF in the aligned variables to obtain the cancellation ideal and a provisional $\boldsymbol{\rho}^{(0)}$;
- (iii) restore $\mathcal{U}$ and every occupied layer of the original δ -dependent $\mathcal{F}$, and determine the corrected $\boldsymbol{\rho}$ from the ideal-jet hierarchy;
- (iv) compose the edge exponents according to $\mathbf{v}_{\text{HR}} = \phi + \boldsymbol{\rho}$, append the final component one, and verify the resolved lower facet and scalefulness.

2.3.2 Boundary hidden regions

An interior HR has all $x_e > 0$. A boundary HR occurs on a contraction stratum

$$x_{e_1} = \cdots = x_{e_k} = 0. \quad (2.39)$$

The graph polynomial is first restricted to that stratum, the vanished variables are removed, and the complete analysis is repeated with every remaining active parameter positive. In graph language this is the corresponding contraction minor; in Landau language it is a subleading Landau singularity.

A complete search must include the relevant contraction strata. Failure to find an interior HR is not a proof that the graph has no HR.

If a contraction stratum also requires asymptotic-order alignment, the restriction is made first and the alignment faces are then constructed from the surviving support. The resulting $\mathcal{F}_\star$ is passed to the core HRF construction. Thus boundary restriction and order alignment are independent wrappers which may be nested without modifying the core logic.

2.4 Interpretation and search-level certification

Once a candidate has passed the core certification, one may reconstruct its momentum-space interpretation. At the level of a systematic search one must also identify equivalent presentations and distinguish exhaustive absence certificates from truncated negative searches.

2.4.1 From parameter scaling to momentum flow

A parameter-space vector fixes propagator virtualities, but it does not by itself fix a unique routing or the separate light-cone components of every edge momentum. We therefore distinguish an *edge-flow assignment* from a *mode-complete momentum region*. The former records the powers of all propagator momenta at a candidate pinch. The latter additionally specifies a basis of independent loop fluctuations and the widths of their integration domains.

Choose a light-cone frame and write

$$q_e \sim (\delta^{a_e}, \delta^{b_e}, \delta^{c_e}) \equiv (q_e^+, q_e^-, |q_{e\perp}|). \quad (2.40)$$

If the certified LP vector is $\mathbf{v}$, inverse Schwinger scaling gives the relative virtuality powers

$$q_e^2 \sim \delta^{\kappa_e}, \quad \kappa_e = \kappa_{\text{ref}} - v_e, \quad (2.41)$$

where one known physical virtuality fixes κ_{ref} . For an ordinary direct realization,

$$\kappa_e = \min(a_e + b_e, 2c_e). \quad (2.42)$$

When $a_e + b_e = 2c_e < \kappa_e$, the longitudinal and transverse terms in $q_e^2 = q_e^+ q_e^- - q_{e\perp}^2$ must cancel at leading order. Such a near-on-shell enhancement requires a coefficient-level check and cannot be certified by valuations alone.

At every vertex and for every component, momentum conservation requires the smallest exponent in the signed sum to occur at least twice. This tropical condition is necessary, but by itself is not sufficient: it does not test the signs and leading coefficients. Our reconstruction therefore proceeds as follows.

Ordinary facet regions. For a facet region the working assumptions, supported by the standard soft and collinear examples, are:

- (i) use the facet vector and an exact external-momentum chart to determine the target κ_e and external component powers;
- (ii) solve Eq. (2.42) together with tropical vertex conservation for every edge;
- (iii) prefer branches in which the largest possible number of propagators realize their virtuality directly and homogeneously, $a_e + b_e = 2c_e = \kappa_e$;
- (iv) choose independent loop variables as chords of a spanning tree, or an equivalent affine rerouting by external momenta, and verify that their component scalings reproduce the complete edge table;
- (v) power count with the fluctuation widths of these independent loop variables, not with a selected subset of edge momenta.

For the facet examples considered below, generic leading coefficients then realize the valuation solution and no additional cancellation-domain support is required.

Hidden regions. For an HR, the same steps are necessary but not sufficient. The emerging picture is:

- (i) use the certified *total* vector, including asymptotic-order alignment, to determine the virtuality powers;
- (ii) construct a complete edge-flow table satisfying virtualities and vertex conservation;
- (iii) impose the cancellation ideal, or equivalently local coordinates transverse to the positive pinch, at coefficient level. This selects the correlated branch and fixes the restricted support that valuations alone cannot see;
- (iv) search all independent loop bases and affine reroutings. A loop mode is Glauber only when an independent integration variable has $a + b > 2c$, so that its transverse virtuality dominates. A transverse-dominated *value* of a combination is evidence for a Glauber direction, but is not yet its fluctuation measure;
- (v) for each potentially restricted light-cone component, express every Feynman denominator in the adapted loop basis and retain those whose dependence on the local fluctuation is leading. Their $i0$ prescriptions determine whether poles approach from opposite half-planes and fix the fluctuation width;
- (vi) determine the remaining independent loop-component widths in the local pinch coordinates and compare the resulting momentum-space power count with the parameter-space count.

The last comparison is a strong certificate because the cancellation depth appears as restricted integration support in both representations. We shall call a result *mode complete* only after this width information is known. A tropical edge-flow table without the coefficient and width checks is a necessary momentum-space candidate, not a proof of the region.

Concretely, suppose a propagator is linear in the selected component,

$$D_e = A_e \ell^+ + B_e + i0, \quad \ell_e^+ = -\frac{B_e}{A_e} - \frac{i0}{A_e}. \quad (2.43)$$

Two active propagators with opposite signs of A_e place their poles on opposite sides of the real axis. If $A_e \sim \delta^{\alpha_e}$ and $D_e \sim \delta^{\kappa_e}$, a leading pole restricts the local coordinate to

$$\Delta \ell^+ \sim \delta^{\kappa_e - \alpha_e}. \quad (2.44)$$

Propagators for which $\alpha_e + \text{pow}(\Delta \ell^+) > \kappa_e$ depend on ℓ^+ algebraically but are subleading in this local fluctuation and do not generate its leading pinch. This is the precise distinction between a central edge valuation and a loop integration width.

2.4.2 When do two presentations describe the same HR?

Generator lists are not physical labels. Two presentations describe the same HR when they determine the same physical component of the common pinch locus, the same resolved

leading facet, and the same pulled-back total scaling vector. Replacing a generator set by another presentation of the same relevant ideal does not create a new region.

Genuinely distinct HRs can arise from:

- different positive components of the cancellation variety;
- different cancellation ideals or singular hypersurface systems; or
- different lower facets in local coordinates, and hence different certified pullback vectors.

Thus a raw multiplicity of accepted generator presentations must be reduced by ideal, positive-locus and final-facet equivalence before being interpreted as multiple physical HRs.

2.4.3 Generator-independent certificates of absence

Factor harvesting is efficient for discovery, but a definitive negative statement should not depend on a maximum factor length, a maximum number of generators or a preferred factorised presentation.

The homogeneity condition (2.11), together with strict separation from all non-SL monomials, implies that the exponent support of $\mathcal{F}_{\text{SL}}$ is an exposed face of the appropriate starting support. This gives a generator-independent route:

1. enumerate every relevant exposed face of $\mathcal{F}_0$, or of the augmented full support when asymptotic-order alignment is allowed;
2. interpret each face polynomial as a candidate $\mathcal{F}_{\text{SL}}$ selected by its face normal, and discard faces that fail elementary necessary conditions;
3. solve the simultaneous positive logarithmic Landau equations for every surviving candidate $\mathcal{F}_{\text{SL}}$ in $\mathcal{K}$;
4. test each positive solution against $\mathcal{U}$, every occupied restored δ -layer, the hierarchy gap and the resolved lower facet;
5. repeat the analysis on all required contraction strata.

If every face is excluded by one of these exact steps, there is no HR within the stated first-sheet and asymptotic assumptions. If a search limit, timeout or undecided positivity question is reached, the result is *unresolved*, not evidence for absence.

This distinction should be visible at the user interface. An *exploratory profile* may impose finite budgets on the number of candidate generator sets or generator unions and on positivity solvers; it is useful for finding constructive witnesses, but cannot certify absence. A *certified profile* removes every correctness-sensitive combinatorial cap and records whether factor construction, candidate enumeration, positivity tests and exact scaling all completed. A no-HR statement is made only when none of these stages was truncated or timed out. Arbitrary signed two-monomial subsums are not part of the complete construction: the latter uses the irreducible factors of the full derivative and primitive kinematic-channel polynomials. Such monomial-pair tests may be retained as a legacy diagnostic, but not as a completeness criterion.

2.5 Algorithm summary

The complete conceptual workflow is: steps 1–3 configure the physical problem and its optional wrappers, steps 4–8 apply the two parts of the core HRF construction, and steps 9–10 manage equivalence and completeness at search level.

1. **Fix the physical chart.** Specify $\delta \rightarrow 0^+$, a common set of real kinematic coordinates $\mathbf{s}$, and the inequalities defining the physical domain $\mathcal{K}$. Introduce positive coordinates only locally if they simplify a particular positivity test.
2. **Choose a stratum.** Start in the interior and repeat on the required contraction strata.
3. **Choose the starting face.** Use $\mathcal{F}_0$ in the ordinary case; when necessary enumerate order-aligned exposed faces $\mathcal{F}^{[\phi]}$ of the full δ -dependent polynomial.
4. **Generate candidate cancellation structures.** Form the derivatives of $\mathcal{F}_\star$ and apply the mixed-sign test only as a necessary prefilter. Harvest primitive non-monomial factors directly whenever literal factorisation or kinematic-sector decomposition exposes them. Do not solve the simultaneous derivative equations of $\mathcal{F}_\star$, nor saturate their full ideal: they generically have no positive solution at this stage when an obstruction is present.
5. **Construct the candidate decomposition.** Use the candidate generator ideal together with the homogeneous-weight and strict-separation conditions to select $\mathcal{S}_{\text{SL}}$ from the starting support. Define $\mathcal{F}_{\text{SL}} = \sum_{i \in \mathcal{S}_{\text{SL}}} m_i$, determine the candidate relative hierarchy, and decompose $\mathcal{F}_\star = \mathcal{F}_{\text{SL}} + \mathcal{F}_{\text{Obs}}$. Retain every intermediate occupied layer for the full certification.
6. **Apply the first-sheet pinch test to $\mathcal{F}_{\text{SL}}$.** Only after $\mathcal{S}_{\text{SL}}$ has been isolated, solve $\partial \mathcal{F}_{\text{SL}} / \partial x_e = 0$ simultaneously for all active parameters with $\mathbf{x} > 0$ and $\mathbf{s} \in \mathcal{K}$. The resulting common positive locus defines the physical cancellation ideal; reject the candidate if this system has no solution. When useful, complete these candidate-specific equations, or explicitly selected derivative-sector relations, by the saturation (2.21).
7. **Resolve all occupied cancellation layers.** Use the ideal filtration of the complete $\mathcal{P} = \mathcal{U} + \mathcal{F}$, positive transverse weights and the actual δ -weight lattice to find the first nonvanishing layer.
8. **Fix and certify the total vector.** Determine $\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1)$, W_{SL} , W_{HR} and the gap. Certify the lower facet in original coordinates when possible; otherwise dissect and pull the facet normal back.
9. **Identify physical equivalence classes.** Merge generator presentations that describe the same positive pinch component and final facet.
10. **State completeness honestly.** A positive HR requires all certification stages. A no-HR conclusion requires exhaustive face and boundary coverage; incomplete searches remain unresolved.

3 Hidden Region Finder: Applications

Colour convention for the examples

Every example will display the relevant part of $\mathcal{F}$ using one colour convention. The colours refer to weights under the *final total region vector*; in a composite limit this includes both the preliminary asymptotic-order alignment and the relative HRF scaling:

$\text{red} = \text{individually superleading terms at } W_{\text{SL}},$ $\text{blue} = \text{resolved leading terms at } W_{\text{HR}},$ $\text{black} = \text{all other terms.}$	(3.1)
--	-------

Thus a conventional facet region contains blue leading terms and black subleading terms, but no red sector. In a hidden region the red monomials have a lower individual weight than the blue monomials, but their sum vanishes on the positive pinch. The blue terms therefore give the actual leading polynomial. Black also includes any intermediate layer that lies in the cancellation ideal and vanishes before the blue layer is reached.

Schematically, the comparison is

$$\mathcal{F}^{(R)} = \delta^{W_R} \mathcal{F}_{W_R} + \mathcal{O}(\delta^{>W_R}), \quad \text{ordinary facet region,} \quad (3.2)$$

$$\mathcal{F}^{(\text{HR})} = \delta^{W_{\text{SL}}} \mathcal{F}_{\text{SL}} + \delta^{W_{\text{HR}}} \mathcal{F}_{\text{HR}} + \mathcal{O}(\delta^{>W_{\text{HR}}}), \quad \text{hidden region,} \quad (3.3)$$

where the first term in (3.3) vanishes on the pinch. This is a classification of the terms, not an instruction to discard the red polynomial before the cancellation has been established.

Every family below is presented in the same order:

- (i) the kinematic expansion and its physical domain;
- (ii) the graph and the assignment of Schwinger parameters;
- (iii) the decomposition $\mathcal{F}_\star = \mathcal{F}_{\text{SL}} + \mathcal{F}_{\text{Obs}}$ and its positive cancellation locus;
- (iv) the certified total vector $(\mathbf{v}; 1)$, including $(W_{\text{SL}}, W_{\text{HR}})$;
- (v) the feature that the algorithm must retain in order not to miss the region.

The examples are therefore a tour through possible HR mechanisms, rather than a chronology of how the implementation was developed.

3.1 Four-point integrals: the Crown seed

The four-point applications are organized by the three-loop Crown. We first identify its interior wide-angle HR, then follow the same seed onto contraction strata of higher-loop graphs, complete the converse No-Crown audit, and finally compare the resulting topology classification with the three Regge channels.

3.1.1 Wide-angle kinematics and the three-loop seed

Take four external momenta in the physical $2 \rightarrow 2$ domain

$$s_{12} > 0, \quad s_{23} < 0, \quad s_{12} > -s_{23}, \quad (3.4)$$

and regulate the on-shell limit by $p_i^2 = \delta q_i^2$, with fixed nonzero q_i^2 and $\delta \rightarrow 0^+$. A common normalization of the q_i^2 is immaterial for the region vector. The Crown has internal edge set

$$E_C = \{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}, \quad (3.5)$$

in the order $x_1, \dots, x_8$, with p_i attached to vertex i for $i = 1, \dots, 4$. This edge list fixes the graph without relying on a particular drawing.

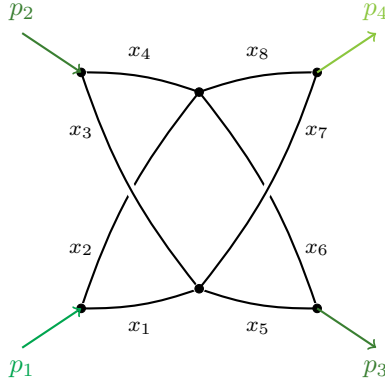


Figure 1. The three-loop Crown with the Schwinger-parameter assignment used in Eq. (3.5). The external-arrow colours are visual label guides only.

At fixed parameters the leading polynomial is

$$\begin{aligned} \mathcal{F}_0 = & s_{12}(x_4x_5 - x_3x_6)(x_2x_7 - x_1x_8) \\ & - s_{23}(x_2x_3 - x_1x_4)(x_6x_7 - x_5x_8). \end{aligned} \quad (3.6)$$

Regard the graph as four two-edge paths between vertices 5 and 6. Factoring one positive Schwinger scale from each path leaves the four ratios

$$r_A = \frac{x_1}{x_2}, \quad r_B = \frac{x_3}{x_4}, \quad r_C = \frac{x_5}{x_6}, \quad r_D = \frac{x_7}{x_8}. \quad (3.7)$$

Then

$$\begin{aligned} \mathcal{F}_0 &= x_2x_4x_6x_8 \Psi_C(\mathbf{r}), \\ \Psi_C(\mathbf{r}) &= s_{12}(r_C - r_B)(r_D - r_A) - s_{23}(r_B - r_A)(r_D - r_C) = \frac{1}{2} \mathbf{r}^T K_C \mathbf{r}, \end{aligned} \quad (3.8)$$

$$K_C = \begin{pmatrix} 0 & s_{12} & -s_{12} - s_{23} & s_{23} \\ s_{12} & 0 & s_{23} & -s_{12} - s_{23} \\ -s_{12} - s_{23} & s_{23} & 0 & s_{12} \\ s_{23} & -s_{12} - s_{23} & s_{12} & 0 \end{pmatrix}.$$

The ratio Landau equations are $K_C \mathbf{r} = 0$. Every row of K_C sums to zero, while in the domain (3.4) a principal three-by-three minor is nonzero:

$$\det(K_C)_{1\dots 3, 1\dots 3} = -2s_{12}s_{23}(s_{12} + s_{23}) \neq 0. \quad (3.9)$$

Consequently K_C has rank three and its complete stationary locus is

$$\mathbf{r} = \rho(1, 1, 1, 1), \quad \rho > 0, \quad \Psi_C(\mathbf{r}) = 0. \quad (3.10)$$

The last equality follows directly from $\Psi_C = \mathbf{r}^T K_C \mathbf{r} / 2$. Thus a positive stationary family exists at every generic point of the physical wide-angle domain; no additional kinematic condition is required. This ratio change is the rescaling step that precedes a local dissection of the Crown pinch. It turns each binomial factor in (3.6) into a positive path scale times a linear difference of two ratio points. Their common zero is $r_A = r_B = r_C = r_D$, which is fixed by only three relative distances, for example $r_B - r_A$, $r_C - r_A$ and $r_D - r_A$. The four displayed binomial constraints therefore contain only three independent equations. Hence

$$\mathcal{F}_\star = \mathcal{F}_0 = \mathcal{F}_{\text{SL}}, \quad \mathcal{F}_{\text{Obs}} = 0. \quad (3.11)$$

Writing $\mathcal{F} = \mathcal{F}_0 + \delta\mathcal{F}_1 + \dots$, the certified vector and weights are

$$\vec{v}_{\text{HR}} = (-1, -1, -1, -1, -1, -1, -1, -1; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-4, -3). \quad (3.12)$$

Consequently

$$\mathcal{P}(\delta^{-1}\mathbf{x}, \delta) = \delta^{-4} \mathcal{F}_0 + \delta^{-3} (\mathcal{F}_1 + \mathcal{U}) + \mathcal{O}(\delta^{-2}). \quad (3.13)$$

The algorithmic lesson is the most elementary one: HRF must keep a simultaneous positive zero of several cancellation factors. No obstruction, non-uniform scaling or preliminary alignment is needed.

3.1.2 Boundary descendants: SuperCrown and HyperCrown

Higher-loop graphs inherit the seed on contraction strata. In edge order, the two graphs are

$$\begin{aligned} E_{SC} &= \{(1, 5), (1, 6), (2, 7), (2, 6), (3, 5), (3, 8), (4, 7), (4, 8), (5, 7), (6, 8), (7, 8), (5, 6)\}, \\ E_{HC} &= \{(1, 5), (1, 6), (4, 9), (4, 6), (2, 7), (2, 6), (3, 8), (3, 6), (9, 5), (7, 8), (8, 9), (5, 7)\}. \end{aligned} \quad (3.14)$$

In this subsection the displayed graphs use the one-based order $x_1, \dots, x_{12}$. The accompanying notebooks use $x_0, \dots, x_{11}$, so that $x_j^{\text{notebook}} = x_{j+1}^{\text{here}}$. We give both conventions when specifying a boundary; this prevents a contraction label from being confused with a scaling component.

SuperCrown: a codimension-two inherited Crown. The focused boundary is

$$B_{\text{SC}} = \{x_9 = x_{10} = 0\}_{\text{here}} = \{x_8 = x_9 = 0\}_{\text{notebook}}. \quad (3.15)$$

After restriction, the surviving superleading polynomial is not merely the Crown polynomial: it includes the two additional active edge parameters,

$$\mathcal{F}_{\text{SL}}^{\text{SC}} = x_{11}x_{12} \mathcal{F}_{\text{SL}}^C(x_1, \dots, x_8), \quad \mathcal{F}_{\text{Obs}}^{\text{SC}} = 0. \quad (3.16)$$

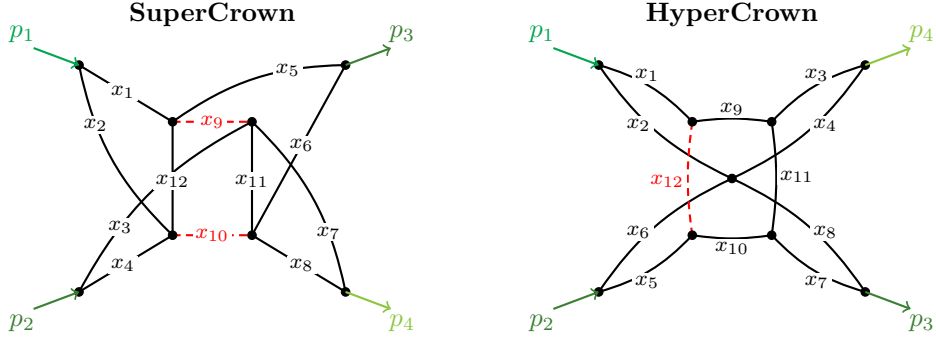


Figure 2. Higher-loop Crown descendants: the five-loop SuperCrown (left) and the four-loop HyperCrown (right). Red dashed edges are set to zero on the representative contraction strata used below: $x_9 = x_{10} = 0$ for the SuperCrown and $x_{12} = 0$ for the HyperCrown, in the one-based convention of the figure.

Equivalently, its two generators are

$$g_1 = (x_4x_5 - x_3x_6)(x_2x_7 - x_1x_8), \quad (3.17)$$

$$g_2 = (x_2x_3 - x_1x_4)(x_6x_7 - x_5x_8), \quad (3.18)$$

with the same simultaneous positive component as the Crown. In the active order $(x_1, \dots, x_8, x_{11}, x_{12})$, the certified augmented vector is

$$\vec{v}_{\text{HR}}^{\text{SC}} = (-1, -1, -1, -1, -1, -1, -1, -1, -1, -1; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-6, -5). \quad (3.19)$$

The weights include the monomial factor $x_{11}x_{12}$; stripping it would incorrectly quote the Crown weights $(-4, -3)$. The current polynomial-factor run finds one HR on this stratum with no search truncation.

HyperCrown: two boundary-HR types. For the HyperCrown the four edges

$$Q_{\text{HC}} = \{x_9, x_{10}, x_{11}, x_{12}\}_{\text{here}} = \{x_8, x_9, x_{10}, x_{11}\}_{\text{notebook}} \quad (3.20)$$

form a square with dihedral symmetry D_4 . External, edge and kinematic labels are permuted together. Consequently one direct positive witness establishes its entire orbit, including crossing-related representatives. The wide-angle results through codimension two are

Type	Representative boundary	Certified scaling	Full orbit / conclusion
I	$x_{12} = 0$	the opposite edge x_{11} has $v_{11} = -2$; all other active $v_e = -1$; $(W_{\text{SL}}, W_{\text{HR}}) = (-6, -5)$	$x_9 = 0, x_{10} = 0, x_{11} = 0, x_{12} = 0$: one HR on each boundary
II	$x_9 = x_{11} = 0$	all ten active components have $v_e = -1$; $(W_{\text{SL}}, W_{\text{HR}}) = (-5, -4)$	$\{x_9, x_{11}\}, \{x_{10}, x_{11}\}, \{x_{10}, x_{12}\}, \{x_9, x_{12}\}$: one HR on each boundary
Excluded	$x_9 = x_{10} = 0$	no exact coverage vector	the opposite-pair orbit $\{x_9, x_{10}\}, \{x_{11}, x_{12}\}$ has no HR

Here every boundary equation means $x_e = 0$; it must not be read as a value of v_e . Type I has the decomposition

$$\mathcal{F}_0^{\text{HC}}|_{x_{12}=0} = \underbrace{[\mathcal{F}_0^{\text{HC}}]_{W=-6}}_{\mathcal{F}_{\text{SL}=x_{11}}^{\text{I}} Q_{\text{I}}} + \underbrace{[\mathcal{F}_0^{\text{HC}}]_{W=-5}}_{\mathcal{F}_{\text{Obs}}^{\text{I}}}, \quad (3.21)$$

where the bracket selects monomials by the displayed Type-I vector. The positive zero of Q_{I} , together with its derivative conditions, is the Landshoff cancellation locus; the obstruction fixes the extra unit of scaling on the opposite square edge. For Type II every surviving term of $\mathcal{F}_0^{\text{HC}}$ has weight -5 , so

$$\mathcal{F}_0^{\text{HC}}|_{x_9=x_{11}=0} = \mathcal{F}_{\text{SL}}^{\text{II}}, \quad \mathcal{F}_{\text{Obs}}^{\text{II}} = 0. \quad (3.22)$$

The two accepted generator presentations recorded for a direct Type-II representative give the same positive component and the same vector; they are one physical HR, not two. The momentum reconstruction gives the complementary interpretation. Type I is the four-jet Landshoff flow with one additional soft loop carried by the opposite square edge. Type II instead adds one loop collinear to an external direction. Neither type requires a generic-angle Glauber mode.

The opposite-pair exclusion is stronger than a failed capped search. On the notebook boundary $x_8 = x_9 = 0$, complete factorisation of all relevant derivatives and primitive kinematic-channel polynomials produces one s_{12} -supported Crown-like generator and no s_{23} -supported companion. Exact coverage rejects the remaining one-generator presentation, with no candidate, union, polynomial-size or solver cap. Its D_4 image excludes $x_{10} = x_{11} = 0$. By contrast, the HyperCrown interior is still unresolved: the completed capped trials contain no HR, but the cap and a timeout prohibit an absence claim.

The family therefore adds two indispensable capabilities. First, the search must descend to contraction strata: an interior-only analysis misses regions present in the parent integral. Second, constructive witnesses and negative statements require different standards. Symmetry can complete an existence orbit, whereas absence requires a complete polynomial-factor and exact-scaling audit with every search budget exposed.

3.1.3 The four-loop No-Crown converse

The converse wide-angle audit completes the four-loop picture. Sixteen mixed-sign topologies without a Crown contraction minor have no interior or boundary HR. The search is complete on the stated graph sample, so these are no-HR certificates rather than failures to find a candidate. Together with the positive Crown descendants above, this establishes both directions of the Crown-minor criterion for the audited wide-angle topologies through four loops.

3.1.4 The Crown family in Regge limits: an audited correspondence

The Regge comparison becomes meaningful only after both sides of the wide-angle classification are visible. Momentum conservation gives $s_{13} = -s_{12} - s_{23}$, and the three channel

charts used in the audit are

$$\begin{aligned}
T_{23} : \quad s_{23} &= -\delta s_{12}, & s_{12} &> 0, \\
T_{12} : \quad s_{12} &= -\delta s_{23}, & s_{23} &< 0, \\
T_{13} : \quad s_{23} &= -s_{12} + \delta s_{12}, & s_{12} &> 0,
\end{aligned} \tag{3.23}$$

with $\delta \rightarrow 0^+$. In the ratio coordinates of Eq. (3.7), the leading Crown polynomial in each chart is, up to a nonzero kinematic and positive monomial factor,

$$\begin{aligned}
\Psi_C^{T_{23}} &\propto (r_C - r_B)(r_D - r_A), \\
\Psi_C^{T_{12}} &\propto (r_B - r_A)(r_D - r_C), \\
\Psi_C^{T_{13}} &\propto (r_C - r_A)(r_D - r_B).
\end{aligned} \tag{3.24}$$

The corresponding positive cancellation loci are therefore

$$\begin{aligned}
T_{23} : \quad r_B &= r_C, \quad r_A = r_D, \\
T_{12} : \quad r_A &= r_B, \quad r_C = r_D, \\
T_{13} : \quad r_A &= r_C, \quad r_B = r_D.
\end{aligned} \tag{3.25}$$

This exposes an important change in codimension. At generic wide angle the Crown locus identifies all four ratio points and hence has three independent relative-distance equations. In each Regge chart the leading polynomial reduces to one product of two difference factors, leaving only two independent equations. Each Regge locus is therefore larger and contains the wide-angle diagonal $r_A = r_B = r_C = r_D$.

Nevertheless, the reduction from three equations to two does not produce new seed topologies in the completed low-loop survey. The positive side is inherited by every audited graph that exposes a Crown contraction minor: the Crown itself has an HR in all three channels, and its higher-loop descendants carry the corresponding region on the contraction stratum. On the negative side, the same sixteen four-loop No-Crown topologies were checked in T_{23} , T_{12} and T_{13} . All 48 interiors and all 343356 nonempty contraction strata through $E - 2$ were resolved, with no HR.

Thus, after the channel relabellings implied by crossing, the set of topologies with a Regge HR agrees exactly with the set having a wide-angle Landshoff HR at three and four loops. Equivalently, in this complete audited sample both are precisely the topologies containing the Crown as a contraction minor. We call this the wide-angle-to-Regge conjecture when extrapolated beyond four loops. Its validity is not a trivial consequence of the Landau equations: algebraically the Regge pinch requires fewer conditions and could have admitted additional topologies, but none occurs in the completed survey.

3.2 Five-point integrals: the fish seed

The examples below use the same unlabeled two-loop six-propagator topology but different external attachments and different kinematic expansions. We refer to this topology as the fish seed. The first is the spacelike-collinear limit for which this seed was originally

identified. The second is an exactly on-shell approach to the planar boundary of five-particle wide-angle scattering. The cancellation ideals and total vectors are different, but in both cases the same sparse topology supports an HR. A central-soft multi-Regge limit reaches the normalized planar boundary by a different route and supplies a third certificate. The three cancellation mechanisms should not be identified merely because the unlabeled graph is common. Before applying the seed to near-planar scattering, we define the underlying massless five-point kinematics separately, since that construction is independent of the graph and its loop order.

3.2.1 The spacelike-collinear limit

The spacelike-collinear application of HRF was presented in the recent short publication [1]. We use there, and here, the kinematic parametrisation introduced in Ref. [2], which originates from a momentum-twistor construction. It makes the dependence on the transverse collinear variable δ especially transparent: the deformation away from the strict collinear surface enters linearly through $c\delta$, while the vanishing collinear invariant starts at $s_{23} = -4\delta^2$. Thus the Mandelstam invariants are polynomial, and at most quadratic, in the expansion variable. In our notation the collinear chart is

$$\begin{aligned} s_{12} &= sz, & s_{23} &= -4\delta^2, & s_{34} &= -s\chi(z-1), \\ s_{45} &= s, & s_{15} &= s\chi + c\delta, & \delta &\rightarrow 0^+, \end{aligned} \quad (3.26)$$

with

$$s > 0, \quad z > 1, \quad -1 < \chi < 0, \quad (3.27)$$

and fixed nonzero c . Keeping χ negative is useful: the same variables then display directly the change from timelike to spacelike collinear kinematics.

The two-loop seed has edge set

$$E_5 = \{(1, 3), (1, 5), (2, 3), (2, 5), (3, 4), (4, 5)\} \longleftrightarrow (x_1, \dots, x_6), \quad (3.28)$$

and external attachment $p_i \rightarrow i$, $i = 1, \dots, 5$. We use $x_{i,j,\dots} = x_i x_j \dots$.

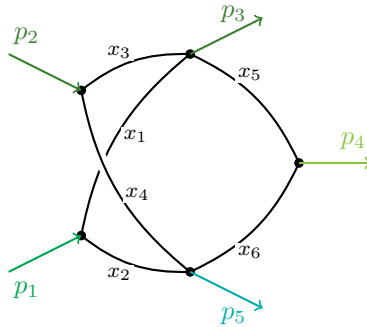


Figure 3. The two-loop five-point seed graph, redrawn in the style of Ref. [1], with the Schwinger parameters used in Eqs. (3.30)–(3.34).

The ordinary facet and the hidden region. For the ordinary facet vector

$$\vec{v}_C = (-2, 0, -2, -2, -2, 0; 1), \quad (3.29)$$

the expanded polynomial is classified as

$$\mathcal{F}_0^{[C]} = s(x_{2,5} + \chi x_{1,6})[(z-1)x_3 - x_4] + sz(\chi+1)x_{2,3,6}, \quad (3.30)$$

$$\mathcal{F}_\delta^{[C]} = -c\delta x_{1,4,6} + c\delta x_{2,3,6} + 4\delta^2 x_{1,4,5} - 4\delta^2 x_{2,3,5}. \quad (3.31)$$

There is no red sector: the blue monomials form an ordinary lower facet.

For the hidden-region vector

$$\vec{v}_{\text{HR}} = (-2, -1, -2, -2, -2, -1; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-5, -4), \quad (3.32)$$

the identical polynomial instead reads

$$\mathcal{F}_0^{[\text{HR}]} = \underbrace{s(x_{2,5} + \chi x_{1,6})[(z-1)x_3 - x_4]}_{\mathcal{F}_{\text{SL}}} + \underbrace{sz(\chi+1)x_{2,3,6}}_{\mathcal{F}_{\text{Obs}}}, \quad (3.33)$$

$$\mathcal{F}_\delta^{[\text{HR}]} = -c\delta x_{1,4,6} + c\delta x_{2,3,6} + 4\delta^2 x_{1,4,5} - 4\delta^2 x_{2,3,5}. \quad (3.34)$$

The cancellation ideal is generated by

$$f_1 = x_{2,5} + \chi x_{1,6}, \quad f_2 = (z-1)x_3 - x_4, \quad \mathcal{I} = \langle f_1, f_2 \rangle. \quad (3.35)$$

The red product vanishes at the positive pinch. The blue obstruction, the blue kinematically suppressed terms, and

$$\mathcal{U}_{W_{\text{HR}}} = x_{1,3} + x_{1,4} + x_{1,5} + x_{3,5} + x_{4,5} \quad (3.36)$$

form the physical layer. This example introduces the obstruction: $\mathcal{F}_{\text{Obs}} \neq 0$ fixes a non-uniform scaling that would not be determined by the factorized cancellation sector alone.

Complete momentum flow. Table 1 gives the edge assignment for both the ordinary facet and the HR. The momentum q_e flows through the edge carrying x_e , and each triple denotes the exponents of $(q_e^+, q_e^-, |q_{e\perp}|)$. The last column in each case is the power κ_e of q_e^2 .

In the facet region the table admits a homogeneous collinear loop basis and no transverse-dominated loop rerouting. In the HR, q_1 and q_5 are separately soft. Vertex conservation forces the minus components to cancel in the independent rerouting

$$\ell_G = q_1 - q_5 \sim (\delta, \delta^2, \delta), \quad (3.37)$$

while one may take $\ell_s = q_1 \sim (\delta, \delta, \delta)$ as the other loop variable. Thus the HR has one soft and one Glauber loop. The entry for q_6 is cancellation enhanced: its leading longitudinal and transverse terms cancel so that $q_6^2 \sim \delta$, despite all three components being of order one. This coefficient-level structure is known explicitly in the spacelike-collinear solution, so this reconstruction is mode complete.

edge	facet $\vec{v}_C$		hidden $\vec{v}_{\text{HR}}$	
	(a_e, b_e, c_e)	κ_e	(a_e, b_e, c_e)	κ_e
q_1	(0, 2, 1)	2	(1, 1, 1)	2
q_2	(0, 0, 1)	0	(1, 0, 1)	1
q_3	(0, 2, 1)	2	(0, 2, 1)	2
q_4	(0, 2, 1)	2	(0, 2, 1)	2
q_5	(0, 2, 1)	2	(1, 1, 1)	2
q_6	(0, 0, 0)	0	(0, 0, 0)	1

Table 1. Complete edge-momentum valuations for the five-point spacelike-collinear facet and HR. Edge q_e corresponds to parameter x_e in Fig. 3.

Parameter- and momentum-space power counting. For unit propagator powers, the scalar LP representation contains $\mathcal{P}^{-D/2}$. The edge-vector sum is $\sum_e v_e = -10$. Only a restricted neighbourhood of the cancellation locus contributes: in local coordinates one cancellation-normal variable has one power less support than a generic variable with the same edge scaling. The parameter measure is therefore

$$\delta^{-10} \delta^{+1} = \delta^{-9}. \quad (3.38)$$

Since the resolved LP polynomial has weight -4 , $\mathcal{P}^{-D/2} \sim \delta^{2D}$, and hence

$$I_{\text{SC}}^{\text{scalar}} \sim \delta^{-9+2D} = \delta^{-1-4\epsilon}, \quad D = 4 - 2\epsilon. \quad (3.39)$$

The power enhancement is a property of the unit-numerator scalar integral. For the gauge-theory numerator relevant to the amplitude, one additional power of δ gives the leading-power behaviour $\delta^{-4\epsilon}$.

The same count is obtained in momentum space. A convenient routing contains one soft loop and one Glauber loop,

$$\ell_s \sim (\delta, \delta, \delta), \quad \ell_G \sim (\delta, \delta^2, \delta), \quad (3.40)$$

where the entries denote $(q^+, q^-, |q_\perp|)$. Their measures scale as δ^D and δ^{D+1} , respectively. The six propagator virtualities have powers

$$(2, 1, 2, 2, 2, 1), \quad (3.41)$$

whose sum is ten. Therefore

$$I_{\text{SC}}^{\text{scalar}} \sim \delta^{D+(D+1)-10} = \delta^{-1-4\epsilon}, \quad (3.42)$$

in agreement with Eq. (3.39). In the original edge routing the Glauber momentum is the difference of two soft edge momenta. This is why inspection of individual propagator modes alone does not reveal it.

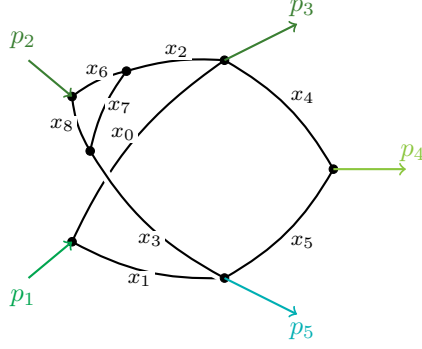


Figure 4. The three-loop vertex-corrected five-point graph, drawn as the two-loop seed of Fig. 3 with a local vertex correction adjacent to p_2 . In particular, x_7 lies outside the other loops. The labels $x_0, \dots, x_8$ are those used in Eqs. (3.44)–(3.46).

3.2.2 A vertex correction: a genuinely non-binomial generator

The same kinematics applied to the three-loop vertex-corrected graph gives

$$E_{5V} = \{(1, 3), (1, 5), (3, 7), (6, 5), (3, 4), (4, 5), (2, 7), (6, 7), (2, 6)\} \longleftrightarrow (x_0, \dots, x_8). \quad (3.43)$$

Here a pairwise monomial search is insufficient. A convenient primitive generator is $g_1 g_2$, where

$$\begin{aligned} g_1 &= (z - 1)(x_2 x_6 + x_2 x_7 + x_2 x_8) + z x_6 x_7 \\ &\quad - (x_3 x_6 + x_3 x_7 + x_3 x_8 + x_6 x_7 + x_7 x_8), \\ g_2 &= x_1 x_4 + \chi x_0 x_5. \end{aligned} \quad (3.44)$$

The first factor has more than two monomials and cannot be represented by a single equality of a monomial pair. The decomposition is nevertheless as simple at ideal level as in the seed:

$$\mathcal{F}_\star = \underbrace{-s g_1 g_2}_{\mathcal{F}_{\text{SL}}} + \underbrace{-s(1 + \chi)z x_1 x_5 (x_2 x_6 + x_2 x_7 + x_6 x_7 + x_2 x_8)}_{\mathcal{F}_{\text{Obs}}}, \quad (3.45)$$

$$\vec{v}_{\text{HR}} = (-2, -1, -2, -2, -2, -1, -2, -2, -2; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-7, -6). \quad (3.46)$$

The lesson is not merely that long factors may occur. HRF must construct and test the ideal generated by primitive polynomial factors, allow polynomial multipliers in $\mathcal{F}_{\text{SL}}$, and solve the positive pinch only after the scaling law has isolated that sector. A binomial-only or pair-sector-only candidate list would miss this region.

3.2.3 Massless five-point kinematics and the near-planar expansion

Let all five external momenta be exactly massless and keep all adjacent Mandelstam invariants at wide angle, with the physical signs

$$s_{12}, s_{34}, s_{45} > 0, \quad s_{23}, s_{15} < 0. \quad (3.47)$$

Define

$$\Gamma_5 = \det(2p_i \cdot p_j)_{i,j=1}^4 = \epsilon_5^2, \quad \epsilon_5 = 4i \varepsilon(p_1, p_2, p_3, p_4). \quad (3.48)$$

Geometrically, Γ_5 is the signed squared four-volume of the parallelotope spanned by $p_1, \dots, p_4$, up to the conventional factor 2^4 ; equivalently it is $2^4(4!)^2$ times the signed squared volume of the corresponding four-simplex. Its vanishing is therefore the condition that the external momenta become coplanar. For real momenta on the physical non-coplanar sheet, ϵ_5 is purely imaginary and $\Gamma_5 < 0$.

To define the approach to this boundary, use the exact five-point variables introduced in the MRK analysis of Ref. [3]. In the (p_1, p_2) light-cone frame, write the complex transverse momentum as $\mathbf{p} = p^x + ip^y$ and its conjugate as $\bar{\mathbf{p}} = p^x - ip^y$, and define

$$w = -\frac{\mathbf{p}_3}{\mathbf{p}_4}, \quad \bar{w} = -\frac{\bar{\mathbf{p}}_3}{\bar{\mathbf{p}}_4}, \quad X_{34} = \frac{p_3^+}{p_4^+}, \quad X_{45} = \frac{p_4^+}{p_5^+}, \quad Q_4^2 = |\mathbf{p}_4|^2. \quad (3.49)$$

For comparing the two expansions below it is useful to replace the inverse transverse ratios by

$$\zeta = -\frac{1}{w} = \frac{\mathbf{p}_4}{\mathbf{p}_3}, \quad \bar{\zeta} = -\frac{1}{\bar{w}} = \frac{\bar{\mathbf{p}}_4}{\bar{\mathbf{p}}_3}. \quad (3.50)$$

These four dimensionless ratios and the transverse scale Q_4^2 form an exact chart, up to an irrelevant longitudinal boost. Transverse momentum conservation gives $\mathbf{p}_3 = -w\mathbf{p}_4$ and $\mathbf{p}_5 = -(1-w)\mathbf{p}_4$, together with their barred counterparts; on-shellness then fixes the minus components from the plus components. In this chart the parity-odd invariant factorises exactly as

$$\epsilon_5 = s_{12}Q_4^2(w - \bar{w}) = s_{12}Q_4^2\frac{\zeta - \bar{\zeta}}{\zeta\bar{\zeta}}. \quad (3.51)$$

Equations (3.49) and (3.50) describe generic five-point kinematics before either expansion is taken. We shall select two distinct paths from this same generic domain. The near-planar path makes the angular factor $\zeta - \bar{\zeta}$ vanish at fixed Q_4^2 . The central-soft MRK path defined later instead makes Q_4^2 vanish while the ratio $\zeta/\bar{\zeta}$ remains generic. The latter is therefore not obtained by first imposing coplanarity and then taking a soft MRK sublimit. In the physical non-coplanar region $\bar{w} = w^*$. The near-planar wide-angle expansion is defined by

$$\begin{aligned} w &= -\frac{1}{\xi} + \frac{i}{2}\lambda, & \bar{w} &= -\frac{1}{\xi} - \frac{i}{2}\lambda, \\ \zeta &= \frac{\xi}{1 - i\xi\lambda/2}, & \bar{\zeta} &= \frac{\xi}{1 + i\xi\lambda/2}, \\ \lambda &\rightarrow 0^+, & \xi, X_{34}, X_{45}, Q_4^2 &\text{fixed}, \end{aligned} \quad (3.52)$$

where $\xi > 0$. Thus the coplanar boundary is simply $\zeta = \bar{\zeta} = \xi$, or equivalently $w = \bar{w} = -1/\xi$. The minus sign makes the positive first-sheet branch manifest in the formulae below. There is no rapidity hierarchy or soft external momentum. Thus

$$\epsilon_5 = ic_\epsilon\lambda, \quad \Gamma_5 = -c_\epsilon^2\lambda^2, \quad c_\epsilon = s_{12}Q_4^2 > 0. \quad (3.53)$$

This kinematic construction is independent of the loop order and of the integral topology considered below.

3.2.4 The fish seed in near-planar scattering

The same six-edge topology analysed in Sec. 3.2.1 has a second certified HR in the near-planar expansion defined above. The externally labelled graph is not the canonical attachment of Fig. 3. It uses

$$(p_1, p_2, p_3, p_5, p_4) \longrightarrow (1, 2, 3, 4, 5), \quad (3.54)$$

as shown in Fig. 5. The internal vertices have deliberately not been moved: the crossing of the p_4 and p_5 external lines makes the changed attachment visible and is not a vertex.

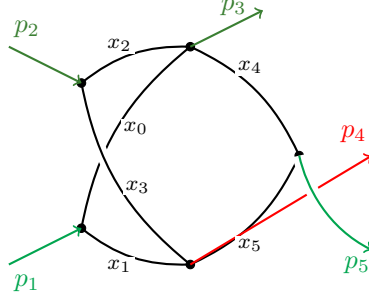


Figure 5. The attachment used for the near-planar and representative central-soft analyses. The internal geometry is identical to Fig. 3, but p_4 and p_5 are interchanged. The colours identify the corresponding external sectors without imposing their geometric directions: (p_2, p_3) are olive, (p_1, p_5) are green, and the central-soft momentum p_4 is red. The edge conventions are $(x_0, \dots, x_5)_{\text{here}} = (x_1, \dots, x_6)_{\text{Fig. 3}}$.

To expose the path symmetry, relabel the six edges as three two-edge paths (x_0, x_1) , (x_2, x_3) and (x_4, x_5) between the two trivalent endpoints, and put

$$r_A = \frac{x_0}{x_1}, \quad r_B = \frac{x_2}{x_3}, \quad r_C = \frac{x_4}{x_5}, \quad \mathcal{F} = x_1 x_3 x_5 \Phi(r_A, r_B, r_C). \quad (3.55)$$

Here $x_1, x_3, x_5 > 0$ are coordinates for the magnitudes on the three two-edge paths, while r_A, r_B, r_C specify the ratios within them; they are together six independent coordinates on the positive Schwinger orthant. Write

$$\Phi(\mathbf{r}) = q_{AB} r_A r_B + q_{BC} r_B r_C + q_{CA} r_C r_A + b_A r_A + b_B r_B + b_C r_C = \frac{1}{2} \mathbf{r}^T K \mathbf{r} + \mathbf{b}^T \mathbf{r},$$

$$K = \begin{pmatrix} 0 & q_{AB} & q_{CA} \\ q_{AB} & 0 & q_{BC} \\ q_{CA} & q_{BC} & 0 \end{pmatrix}, \quad \mathbf{b} = (b_A, b_B, b_C)^T. \quad (3.56)$$

For the external attachment in Eq. (3.54), the entries are Mandelstam invariants, which may be expressed in terms of the adjacent ones using five-point momentum conservation:

$$\begin{aligned} q_{AB} &= s_{35} = s_{12} - s_{34} - s_{45}, & b_A &= s_{14} = -s_{15} + s_{23} - s_{45}, \\ q_{BC} &= s_{13} = -s_{12} - s_{23} + s_{45}, & b_B &= s_{24} = s_{15} - s_{23} - s_{34}, \\ q_{CA} &= s_{23}, & b_C &= s_{45}. \end{aligned} \quad (3.57)$$

Direct differentiation with respect to the ratios gives

$$\nabla_{\mathbf{r}}\Phi = K\mathbf{r} + \mathbf{b}. \quad (3.58)$$

The ratio-stationarity condition at $\mathbf{r} = \boldsymbol{\rho}$ is therefore the logically separate equation

$$K\boldsymbol{\rho} = -\mathbf{b}. \quad (3.59)$$

The determinant of the coefficient matrix is

$$\det K = 2q_{AB}q_{BC}q_{CA}. \quad (3.60)$$

Hence, whenever $q_{AB}q_{BC}q_{CA} \neq 0$, Eq. (3.59) determines a unique ratio vector $\boldsymbol{\rho} = (\rho_A, \rho_B, \rho_C)$. Already at this generic kinematic point, define the three polynomials normal to the ratio-stationary locus,

$$f_A = x_0 - \rho_A x_1, \quad f_B = x_2 - \rho_B x_3, \quad f_C = x_4 - \rho_C x_5. \quad (3.61)$$

Here the $\rho_I = -(K^{-1}\mathbf{b})_I$ retain their full kinematic dependence; no coplanar or MRK limit has yet been imposed.

Ratio stationarity is necessary but is not yet the complete Landau condition. To see the missing equation, use the independent path-magnitude coordinates $u_A = x_1$, $u_B = x_3$, $u_C = x_5$. At fixed ratios,

$$\mathcal{F} = u_A u_B u_C \Phi, \quad \frac{\partial \mathcal{F}}{\partial u_A} = u_B u_C \Phi, \quad \text{and cyclically.} \quad (3.62)$$

Since the path magnitudes are positive, stationarity in these directions also requires

$$\Phi(\boldsymbol{\rho}) = 0. \quad (3.63)$$

In the original Schwinger parameters this condition follows automatically from homogeneity once all derivative equations are imposed. In the ratio chart, however, Φ contains both quadratic and linear terms and is not homogeneous, so Eq. (3.63) does not follow from Eq. (3.59). The stationary value follows directly from Schur's formula for a bordered determinant. Equation (3.59) gives

$$\Phi(\boldsymbol{\rho}) = \frac{1}{2}\mathbf{b}^T \boldsymbol{\rho} = -\frac{1}{2}\mathbf{b}^T K^{-1}\mathbf{b}.$$

For a block matrix, Schur's determinant formula is [4]

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad \det M = \det A \det(D - CA^{-1}B).$$

Taking $M = \mathcal{B}$, $A = K$, $B = \mathbf{b}$, $C = \mathbf{b}^T$, and $D = 0$, one obtains

$$\mathcal{B} = \begin{pmatrix} K & \mathbf{b} \\ \mathbf{b}^T & 0 \end{pmatrix}, \quad \det \mathcal{B} = 2 \det K \Phi(\boldsymbol{\rho}) = 4q_{AB}q_{BC}q_{CA} \Phi(\boldsymbol{\rho}).$$

Using the Mandelstam identifications in Eq. (3.57), five-point momentum conservation gives $\det \mathcal{B} = \det(2p_i \cdot p_j)_{i,j=1}^4 = \Gamma_5$. Consequently,

$$\Phi(\boldsymbol{\rho}) = \frac{\Gamma_5}{4q_{AB}q_{BC}q_{CA}}. \quad (3.64)$$

Completing the quadratic form about $\boldsymbol{\rho}$, and then returning to the original Schwinger parameters, gives the exact identity

$$\mathcal{F} = q_{AB}x_5f_Af_B + q_{BC}x_1f_Bf_C + q_{CA}x_3f_Cf_A + \frac{\Gamma_5}{4q_{AB}q_{BC}q_{CA}}x_1x_3x_5. \quad (3.65)$$

This is an identity for the full massless $\mathcal{F}$ at generic five-point kinematics, not merely its restriction to the Gram surface. It makes direct contact with the HRF derivative strategy, but it is important not to mistake this completed form for the algorithmic starting point. HRF starts from the usual Mandelstam representation of $\mathcal{F}$. For example, direct differentiation there gives

$$D_0 \equiv \frac{\partial \mathcal{F}}{\partial x_0} = q_{AB}x_2x_5 + q_{CA}x_3x_4 + b_Ax_3x_5. \quad (3.66)$$

Literal collection in q_{AB} , q_{CA} and b_A exposes only monomials. The first component of $K\boldsymbol{\rho} = -\mathbf{b}$,

$$b_A = -q_{AB}\rho_B - q_{CA}\rho_C, \quad (3.67)$$

instead reorganises the same derivative as

$$D_0 = q_{AB}x_5(x_2 - \rho_Bx_3) + q_{CA}x_3(x_4 - \rho_Cx_5) = q_{AB}x_5f_B + q_{CA}x_3f_C. \quad (3.68)$$

Thus the simple appearance of the f_I in the following equations has already used the nontrivial stationary reorganisation. In this example the expansion has already been fixed at $\Gamma_5 = 0$, and the complete leading polynomial is the cancellation sector: $\mathcal{F}_\star = \mathcal{F}_0 = \mathcal{F}_{\text{SL}}$. Candidate-specific saturation of its gradient ideal is therefore legitimate and performs this reorganisation without assuming $\boldsymbol{\rho}$ in advance. This special equality is precisely what would fail for a generic starting polynomial containing an obstruction. Since the final term of Eq. (3.65) is independent of x_0, x_2, x_4 , the three stationary-adapted derivatives are

$$\begin{aligned} \frac{\partial \mathcal{F}}{\partial x_0} &= q_{AB}x_5f_B + q_{CA}x_3f_C, \\ \frac{\partial \mathcal{F}}{\partial x_2} &= q_{AB}x_5f_A + q_{BC}x_1f_C, \\ \frac{\partial \mathcal{F}}{\partial x_4} &= q_{BC}x_1f_B + q_{CA}x_3f_A. \end{aligned} \quad (3.69)$$

The determinant of this linear system in (f_A, f_B, f_C) is $2q_{AB}q_{BC}q_{CA}x_1x_3x_5$. Hence, in the interior positive orthant and away from degenerate kinematics, these three derivative equations imply $f_A = f_B = f_C = 0$. The remaining derivatives then reduce to

$$\left(\frac{\partial \mathcal{F}}{\partial x_1}, \frac{\partial \mathcal{F}}{\partial x_3}, \frac{\partial \mathcal{F}}{\partial x_5} \right) \Big|_{f_A=f_B=f_C=0} = \frac{\Gamma_5}{4q_{AB}q_{BC}q_{CA}}(x_3x_5, x_1x_5, x_1x_3). \quad (3.70)$$

Likewise, Eq. (3.65) restricted to this locus equals $\Gamma_5 x_1 x_3 x_5 / (4q_{AB} q_{BC} q_{CA})$. Thus the full derivative system, viewed as a direct Landau analysis before choosing the expansion, first exposes the candidate normal ideal $\langle f_A, f_B, f_C \rangle$, and its remaining equations then require $\Gamma_5 = 0$. This explains why the near-planar expansion is required. The HRF audit below starts only after that expansion has been fixed; on its leading surface the saturation is therefore applied to $\mathcal{F}_{\text{SL}}$, not to a generic obstructed starting polynomial.

Positivity of the resulting ρ_I is precisely the usual interior first-sheet condition $x_e > 0$, because the ρ_I are the stationary values of the positive ratios in Eq. (3.55). The condition $\Gamma_5 = 0$ is the coplanar boundary approached in the near-planar limit, and it is also reached by the central-soft multi-Regge limit discussed below.

This is the precise contrast with the Crown. The Crown matrix K_C in (3.8) has a kinematics-independent zero mode, and its stationary value vanishes throughout the generic wide-angle domain. Here K is generically invertible: it fixes all three ratios, and the remaining Landau equation restricts the kinematics to $\Gamma_5 = 0$. Only now specialise the generic construction to the near-planar expansion. The normals become especially transparent in the exact chart (3.49). On the coplanar boundary put

$$\zeta = \bar{\zeta} = \xi, \quad R = \frac{\xi + X_{45}(\xi + 1)}{1 + X_{34}X_{45}(\xi + 1)}. \quad (3.71)$$

The solution of $K\rho = -\mathbf{b}$ is then

$$\rho_A = X_{34}\xi R, \quad \rho_B = R, \quad \rho_C = \xi, \quad (3.72)$$

and therefore

$$f_A = x_0 - X_{34}\xi R x_1, \quad f_B = x_2 - R x_3, \quad f_C = x_4 - \xi x_5. \quad (3.73)$$

On the physical positive-Landau branch $X_{34}, X_{45}, \xi > 0$, so $R > 0$ and all three stationary ratios are positive. No MRK hierarchy is used in these formulae.

To define the near-planar expansion, let κ denote four coordinates tangent to the Gram surface and use Γ_5 as a local normal coordinate. Then

$$\mathcal{F}(\Gamma_5, \kappa) = \mathcal{F}_0(\kappa) + \Gamma_5 \mathcal{F}_{\Gamma_5}(\kappa) + O(\Gamma_5^2). \quad (3.74)$$

There is no conflict between the apparently nonlinear kinematic dependence in Eq. (3.65) and the linearity of $\mathcal{F}$ in Mandelstam invariants. The Gram determinant Γ_5 is itself a nonlinear polynomial in those invariants. Replacing one Mandelstam variable by Γ_5 is therefore a nonlinear change of kinematic coordinates, and at fixed κ

$$\mathcal{F}_{\Gamma_5} = \left. \frac{\partial \mathcal{F}}{\partial \Gamma_5} \right|_{\kappa} = \sum_a \frac{\partial \mathcal{F}}{\partial s_a} \left. \frac{\partial s_a}{\partial \Gamma_5} \right|_{\kappa}. \quad (3.75)$$

The Jacobian factors on the right need not be linear functions of the remaining kinematics. Equivalently, the separate terms in Eq. (3.65) contain the rational ratios $\rho = -K^{-1}\mathbf{b}$; their nonlinear dependence cancels in the sum, restoring the original Mandelstam-linear polynomial.

The complete coefficient $\mathcal{F}_{\Gamma_5}$ away from the stationary locus depends on the choice of the tangent coordinates κ . Its restriction to $f_A = f_B = f_C = 0$, however, is fixed by Eq. (3.65):

$$\mathcal{F}_{\Gamma_5}|_{f_A=f_B=f_C=0} = \frac{x_1 x_3 x_5}{4q_{AB}q_{BC}q_{CA}}. \quad (3.76)$$

This stationary contribution is not an “obstruction” in the technical HRF sense: it is absent from the leading polynomial by the definition of the chosen kinematic expansion, rather than being made subleading by the HR scaling. Since $\Gamma_5 = O(\lambda^2)$, this contribution first appears two powers beyond $\mathcal{F}_0$. Its precise weight relative to the cancelled leading sector and to $\mathcal{U}$ will follow only after the region vector has been determined below. The leading polynomial itself has the symmetric factorised form

$$\mathcal{F}_0 \equiv \mathcal{F}|_{\Gamma_5=0} = q_{AB}x_5 f_A f_B + q_{BC}x_1 f_B f_C + q_{CA}x_3 f_C f_A. \quad (3.77)$$

Thus the HR generators are the factorised products $f_A f_B, f_B f_C, f_C f_A$, not the individual normal equations. This is a three-generator ideal and is more general than the pair-sector search used in the earlier examples.

Using Eq. (3.57), the outer coefficients in these six summands are the distinct invariant sectors $q_{AB} = s_{35}$, $q_{BC} = s_{13}$ and $q_{CA} = s_{23}$. If the stationary-adapted derivatives in Eq. (3.69) have been obtained, their sector-decomposed form exposes the normals directly, up to positive monomial factors: for example the two sectors of the first line expose f_B and f_C , and the remaining lines expose all three f_I . In the Crown this form is available by direct invariant-sector harvesting. Here the f_I contain kinematic stationary ratios, so the corresponding reorganisation must first be derived. The full Landau analysis above first shows that the remaining equations require $\Gamma_5 = 0$. Once the expansion has been fixed on that surface, saturation by the nonzero determinant following Eq. (3.69) is candidate-specific and the ideal generated by these three leading-sector derivatives is precisely $\langle f_A, f_B, f_C \rangle$. It therefore recovers the three normals without assuming them in advance. Factoring each derivative separately cannot do so in general, because an individual derivative is a linear combination of two normals rather than a multiple of either one. Substitution into $\mathcal{F}_0$ then identifies the HR generator ideal $\langle f_A f_B, f_B f_C, f_C f_A \rangle$.

For the expansion defined in Eqs. (3.52) and (3.53), the certified original-coordinate vector and weights are

$$\vec{v}_{\text{HR}} = (-2, -2, -2, -2, -2, -2; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-6, -4). \quad (3.78)$$

The comparison announced above is now fixed. Every cubic monomial in $\mathcal{F}_0$ has raw weight -6 . The normal factor $\Gamma_5 = O(\lambda^2)$ raises the weight of $\Gamma_5 \mathcal{F}_{\Gamma_5}$ to -4 , while the quadratic polynomial $\mathcal{U}$ also has weight -4 . Cancellation on the singular locus promotes $\mathcal{F}_0$ from $W_{\text{SL}} = -6$ to the same resolved weight $W_{\text{HR}} = -4$. Thus the cancelled leading sector, the first Gram-normal correction and $\mathcal{U}$ all enter the resolved LP polynomial at W_{HR} . The dissection coordinates can be stated explicitly. Take the tangential path magnitudes $u_A = x_1$, $u_B = x_3$, $u_C = x_5$ and the three normals $y_I = f_I$ of Eq. (3.61). In a sign sector

put

$$\begin{aligned} x_0 &= \rho_A u_A + \sigma_A t_A, \quad x_1 = u_A, \quad t_A = |y_A|, \\ x_2 &= \rho_B u_B + \sigma_B t_B, \quad x_3 = u_B, \quad t_B = |y_B|, \quad \sigma_I = \pm 1. \\ x_4 &= \rho_C u_C + \sigma_C t_C, \quad x_5 = u_C, \quad t_C = |y_C|, \end{aligned}$$

At fixed kinematics this transformation has unit absolute Jacobian. The ordinary lower facet in these local coordinates has $u_I \sim \lambda^{-2}$ and $t_I \sim \lambda^{-1}$, and its pullback is Eq. (3.78).

First perform the power count entirely in parameter space and in the original variables. Without the cancellation restriction, six measures with $x_e \sim \lambda^{-2}$ would give λ^{-12} . For each pair, however, only a relative neighbourhood $|f_I| = O(\lambda^{-1})$ contributes, rather than the natural width $O(\lambda^{-2})$. The three restrictions restore one power of λ each. Equivalently, the dissected measure contains three u_I and three t_I integrations. Since the resolved LP polynomial has weight -4 ,

$$\begin{aligned} I_{\text{par}}^{\text{scalar}} &\sim \underbrace{\lambda^{-12}}_{\text{unrestricted}} \underbrace{\lambda^3}_{\text{restricted widths}} \underbrace{(\lambda^{-4})^{-D/2}}_{\mathcal{P}^{-D/2}} \\ &= \underbrace{(\lambda^{-2})^3 (\lambda^{-1})^3}_{\prod_I du_I dt_I \text{ after dissection}} (\lambda^{-4})^{-D/2} = \lambda^{2D-9} = \lambda^{-1-4\epsilon}, \quad D = 4 - 2\epsilon. \end{aligned} \quad (3.79)$$

The momentum-space count is independent. Choose two path momenta, say q_A and q_B , as the loop variables; momentum conservation fixes $q_C = -p_3 - q_A - q_B$. Each unrestricted collinear loop measure scales as λ^D : the momentum fraction has width $O(1)$, its conjugate light-cone component has width $O(\lambda^2)$, and the $D - 2$ transverse components each have width $O(\lambda)$. Keeping the dependent momentum q_C collinear to its wide-angle direction further confines $\delta\xi_A$ and $\delta\xi_B$ to widths $O(\lambda)$ and $O(\lambda^2)$, giving the additional λ^3 . All six propagator virtualities are $q_e^2 = O(\lambda^2)$. Therefore

$$I_{\text{mom}}^{\text{scalar}} \sim \underbrace{(\lambda^D)^2}_{\text{collinear measures}} \underbrace{\lambda^3}_{\text{restricted support}} \underbrace{(\lambda^{-2})^6}_{\text{propagators}} = \lambda^{2D-9} = \lambda^{-1-4\epsilon}. \quad (3.80)$$

The agreement of Eqs. (3.79) and (3.80) checks both the cancellation depth in the original Schwinger domain and its translation into restricted loop-momentum support.

3.2.5 Central-soft MRK: the full-layer certificate

Starting again from the generic chart (3.49)–(3.50), keep the two MRK rapidity gaps large while scaling all components of the central momentum as $p_4^\mu = O(\delta^a)$, with the rapidity parameter $x_{\text{MRK}} = O(\delta^b)$ and $0 < a < b$. The same exact variables used above define this path by

$$\begin{aligned} \zeta &= \widehat{\zeta} \delta^a, & \bar{\zeta} &= \widehat{\bar{\zeta}} \delta^a, & Q_4^2 &= \widehat{Q}_4^2 \delta^{2a}, \\ X_{34} &= \widehat{X}_{34} \delta^{-(a+b)}, & X_{45} &= \frac{\widehat{X}_{45} \delta^{a-b}}{(1 + \widehat{\zeta} \delta^a)(1 + \widehat{\bar{\zeta}} \delta^a)}. \end{aligned} \quad (3.81)$$

All hatted quantities are fixed and nonzero. In the physical region $\widehat{\zeta} = \widehat{\zeta}^*$; the denominator in X_{45} keeps the path exactly on shell rather than specifying only its leading valuation. In

this chart,

$$\epsilon_5 = s_{12} Q_4^2 \frac{\zeta - \bar{\zeta}}{\zeta \bar{\zeta}}. \quad (3.82)$$

Near-planar wide-angle scattering sends $\zeta - \bar{\zeta} \rightarrow 0$ at fixed Q_4^2 , whereas central-soft MRK sends $Q_4^2 \rightarrow 0$ and $\zeta, \bar{\zeta} \rightarrow 0$ together. Both reach a normalized planar boundary, but this kinematic relation does not imply the same HR.

For reference, the earlier executable audit used finite positive coefficients P, K, M, R, T and a transverse coefficient C . It is exactly the same path under

$$\hat{Q}_4^2 = R, \quad \hat{X}_{34} = \frac{P}{K}, \quad \hat{X}_{45} = \frac{KM}{T}, \quad \widehat{\zeta\zeta} = \frac{R}{T}, \quad \hat{\zeta} + \widehat{\bar{\zeta}} = \frac{\sigma C}{T}. \quad (3.83)$$

This map is used only as an exact equivalence and positivity audit below; the common variables in Eq. (3.81) define the expansion.

For the representative central-soft attachment and $(a, b) = (1, 2)$, the alignment face gives

$$\begin{aligned} \phi &= (-3, -3, 0, -3, -3, -3), \\ \mathcal{F}_{-10}^{\text{align}} &= -\frac{\hat{X}_{45} \hat{Q}_4^2}{\widehat{\zeta\zeta}} (\hat{X}_{34} x_2 - x_3)(x_1 x_4 - x_0 x_5). \end{aligned} \quad (3.84)$$

The relative HRF vector is uniform,

$$\rho = (-1, -1, -1, -1, -1, -1), \quad \tilde{v}^{\text{CS}} = (-4, -4, -1, -4, -4, -4). \quad (3.85)$$

The exact chart is Laurent in δ . Factoring out the growing hard invariant s_{12} implements the dimensionless normalization in which Eq. (3.85) is the infrared HRF vector, and

$$\tilde{v}_{\text{HR}}^{\text{CS}} = (-4, -4, -1, -4, -4, -4; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-9, -8). \quad (3.86)$$

Adding +4 uniformly converts to dimensionful Schwinger parameters while $s_{12} \sim \delta^{-4}$ itself varies, but the resulting non-negative list is a change of hard-scale units, not the infrared region vector. The next layer contains

$$-\frac{\hat{Q}_4^2}{\widehat{\zeta\zeta}} x_0 x_3 x_4, \quad (3.87)$$

which remains nonzero on the two-factor locus and has weight -8 , exactly the leading $\mathcal{U}$ weight. In the representative labeling,

$$\mathcal{U}_{-8} = x_0 x_3 + x_1 x_3 + x_0 x_4 + x_1 x_4 + x_3 x_4 + x_0 x_5 + x_1 x_5 + x_3 x_5. \quad (3.88)$$

Thus Eq. (3.87) is not an obstruction below the physical layer; it is part of that layer. Together with the promoted factorised terms, the resolved support has affine rank six and the unique inward normal is precisely Eq. (3.86). The hierarchy gap is one.

The result is stable for the tested rates $(a, b) = (1, 2), (1, 3), (2, 3)$ and for both transverse-sign charts. Among the twelve inequivalent external attachments of the fixed topology,

exactly six have this certificate: those in which the central gluon is attached to either four-valent internal vertex. The other six have no staged candidate. For each positive attachment two additional staged presentations fail the final lower-facet test, so the accepted count is one rather than three.

The reason is already visible on the leading face exposed by the natural alignment vector $\phi = (-3, -3, 0, -3, -3, -3)$. If the soft particle p_4 is emitted from the trivalent middle vertex 4, the aligned face is

$$\mathcal{F}_{\text{mid}}^{\text{align}} = -\frac{\widehat{X}_{45}\widehat{Q}_4^2}{\widehat{\zeta}\widehat{\zeta}}x_1[(\widehat{X}_{34}x_2 - x_3)x_4 + \widehat{X}_{34}x_2x_5]. \quad (3.89)$$

The bracket can vanish for positive parameters, but the zero is not stationary:

$$\frac{\partial}{\partial x_5}[(\widehat{X}_{34}x_2 - x_3)x_4 + \widehat{X}_{34}x_2x_5] = \widehat{X}_{34}x_2 > 0. \quad (3.90)$$

The unmatched term $\widehat{X}_{34}x_2x_5$ records that the soft leg has selected the middle of one particular theta path.

If p_4 is attached instead to the four-valent theta endpoint 5, the corresponding face factorises,

$$\mathcal{F}_{\text{end}}^{\text{align}} = -\frac{\widehat{X}_{45}\widehat{Q}_4^2}{\widehat{\zeta}\widehat{\zeta}}(\widehat{X}_{34}x_2 - x_3)(x_1x_4 - x_0x_5). \quad (3.91)$$

The two factors are independent routing differences. Their simultaneous zero is stationary because $d(AB) = B dA + A dB$. At the endpoint all three theta paths remain available, allowing two independent rebalancings of the two-loop flow. Emission in the middle of one path distinguishes that path and prevents the second rebalancing. The full twelve-attachment scan confirms that no alternative aligned face restores a pinch for a trivalent attachment.

For the scalar power count define the local normals

$$A = x_2 - \widehat{X}_{34}^{-1}\delta^3x_3, \quad B = x_1x_4 - x_0x_5. \quad (3.92)$$

Their generic weights are -1 and -8 , respectively, and the HRF gap restricts A to weight 0. Four tangential coordinates contribute -16 . Solving for x_2, x_5 gives the Jacobian $-1/x_0$, of weight $+4$. The parameter-space measure therefore contributes δ^{-20} , while the resolved LP weight -8 gives $\mathcal{P}^{-D/2} \sim \delta^{4D}$.

In momentum space choose $r = q_0$ and $\ell = q_0 - q_4$. A complete edge assignment is

e	0	1	2	3	4	5	
$(q_e^+, q_e^-, q_{e\perp})$	$(4, 0, 2)$	$(4, 0, 2)$	$(0, 1, 2)$	$(3, 1, 2)$	$(4, 0, 2)$	$(4, 0, 2)$	(3.93)
q_e^2	4	4	1	4	4	4	

where the entries are powers of δ in hard-scale-normalized momenta. The q_0, q_1 poles pinch r^+ with width δ^4 ; q_4, q_5 pinch ℓ^+ with width δ^4 ; and q_2, q_3 pinch ℓ^- with width δ . Both

transverse widths scale as δ^2 . Hence the loop widths are $(4, 0, 2)$ and $(4, 1, 2)$, giving measure power $4D + 1$. The denominator powers sum to 21. The two representations agree:

$$I_{\text{CS}}^{\text{scalar}} \sim \underbrace{\delta^{-20} \delta^{4D}}_{\text{parameter space}} = \underbrace{\delta^{4D+1} \delta^{-21}}_{\text{momentum space}} = \delta^{4D-20} = \delta^{-4-8\epsilon}, \quad D = 4 - 2\epsilon. \quad (3.94)$$

This is the reduced integral with s_{12} factored out, which is the appropriate convention for diagnosing an infrared region. Restoring the engineering-dimension factor s_{12}^{D-6} , with $s_{12} \sim \delta^{-4}$, is a separate overall conversion and must not be used to redefine the HRF vector.

The limiting algebraic structures of the two regions nevertheless have a relation; this comparison does not construct either limit as a sublimit of the other. On the aligned central-soft face the two corresponding path ratios coincide, which may be written $\rho_A = \rho_C$, so

$$x_1 x_4 - x_0 x_5 = x_1 f_C - x_5 f_A, \quad \hat{X}_{34} x_2 - x_3 = \hat{X}_{34} f_B. \quad (3.95)$$

Their product lies in the near-planar ideal generated by $f_A f_B$ and $f_B f_C$; the third generator $f_A f_C$ is suppressed on the selected MRK face. Thus the two limits expose a common path-normal geometry of the same seed topology, but they are not the same region: their physical vectors, momentum modes and scalar powers are different.

3.2.6 Comparison of the certified five-point regions

Expansion	total edge vector	scalar power	momentum signature
Spacelike collinear	$(-2, -1, -2, -2, -2, -1; 1)$	$\delta^{-1-4\epsilon}$	soft + Glauber
Near-planar wide angle	$(-2, -2, -2, -2, -2, -2; 1)$	$\lambda^{-1-4\epsilon}$	three collinear path flows with restricted support
Central-soft MRK	$(-4, -4, -1, -4, -4, -4; 1)$	$\delta^{-4-8\epsilon}$	one on-shell-balanced loop and one Glauber loop

These are not three coordinate descriptions of one region. Their kinematic limits, cancellation ideals, vectors and momentum-space realizations are different. What they share is the same unlabeled two-loop seed. The comparison suggests a broader organisational principle: hidden regions are sparse because only special graph topologies support the required positive pinches, but once such a seed exists it can reappear through distinct cancellation mechanisms in more than one kinematic expansion. The Crown in wide-angle and Regge limits is the four-point example; spacelike-collinear, near-planar wide-angle scattering and central-soft MRK supply three certified five-point examples on an equal footing. The central-soft entry has the two-factor locus in Eq. (3.84); its scalar power in the table is independently reproduced in parameter and momentum space. The six-point NMRK/DSC pair below supplies another example, although its topological interpretation remains to be completed.

3.3 Six-point integrals: the twisted-hexagon seed

The six-point examples are organized by the non-planar external ordering of the one-loop hexagon, which we call the twisted-hexagon seed. We first fix the graph and its attachment

and survey its two-loop relatives, then compare the NMRK surface $w = z$ with the double spacelike-collinear limit.

3.3.1 Common graph and why $\mathcal{F}_0$ is insufficient

Both examples use the one-loop hexagon

$$\begin{aligned} (5, 6), (6, 1), (1, 2), (2, 3), (3, 4), (4, 5) &\longleftrightarrow x_0, x_1, x_2, x_3, x_4, x_5, \\ p_1 \rightarrow 1, \quad p_2 \rightarrow 4, \quad p_3 \rightarrow 5, \quad p_4 \rightarrow 3, \quad p_5 \rightarrow 2, \quad p_6 \rightarrow 6, \end{aligned} \quad (3.96)$$

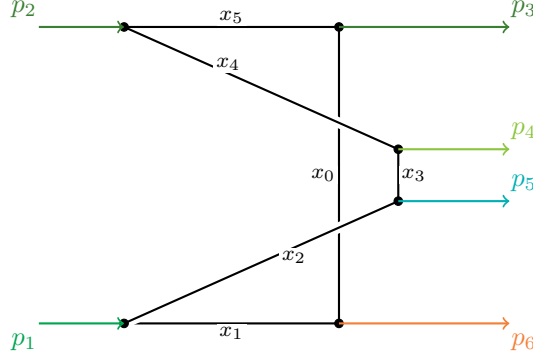


Figure 6. One-loop hexagon which displays a HR in both the (no recoil) NMRK and the DSC limits considered. The nearly straight upper chain $p_2 - x_5 - p_3$ and lower chain $p_1 - x_1 - p_6$ display the two spacelike-collinear pairs. On the right, the vertical separations display the rapidity ordering $p_3 \gg (p_4, p_5) \gg p_6$; the relative order of p_4, p_5 is immaterial in NMRK. The two crossings are deliberate and are not additional vertices. This physical attachment ordering supports the HR (and associated Regge cut), in contrast to the genuinely planar ordering 123456 with all four emissions on one side.

with $\mathcal{U} = x_0 + x_1 + x_2 + x_3 + x_4 + x_5$. The full graph polynomial is the same in both limits; only the invariant substitution changes.

In each case the cancelling monomials occur at different fixed-parameter orders. Thus $\mathcal{F}_0$ does not contain the relevant cancellation sector. A face vector f first performs asymptotic-order alignment and selects $\mathcal{F}_\star = \mathcal{F}^{[f]}$. Because f and $f + c\mathbf{1}$ select the same face, this discovery vector is not yet the physical region vector. The full occupied layer structure and $\mathcal{U}$ must fix the uniform component.

3.3.2 Two-loop topology test and possible Regge-cut interpretation

The two nearly horizontal chains in Fig. 6 are the (23) and (16) collinear sectors. A t -channel exchange is an internal path connecting these two chains. The two central emissions p_4, p_5 may occur on the same such path or on two separate paths. This distinction becomes explicit in the two-loop graphs used in the NMRK and DSC audits. With the external attachments of Eq. (3.96), their internal edges, in $x_0, \dots, x_8$ order, are

$$\begin{aligned} E_{\text{PH}} &= \{(5, 6), (1, 8), (8, 6), (1, 2), (2, 3), (3, 4), (4, 7), (7, 5), (7, 8)\}, \\ E_{\text{NPH}} &= \{(5, 8), (1, 8), (1, 2), (2, 3), (7, 3), (7, 5), (4, 7), (6, 8), (4, 6)\}, \\ E_{\text{HP}} &= \{(5, 6), (1, 8), (8, 6), (1, 3), (3, 4), (4, 7), (7, 5), (7, 2), (8, 2)\}. \end{aligned} \quad (3.97)$$

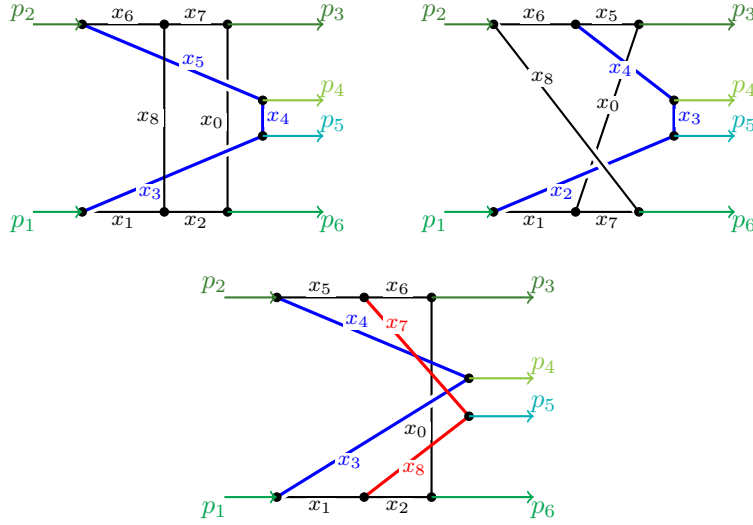


Figure 7. Two-loop graphs, drawn in the same rapidity-ordered embedding as Fig. 6. In each graph the upper and lower horizontal chains are the (23) and (16) collinear sectors. The pair of diagrams at the top correspond to a planar and a nonplanar hexagonbox topologies in which both central emissions emerge from the same t -channel exchange (drawn in blue). The lower graph represents a hexagon-pentagon topology in which the two central emissions p_4 and p_5 emerges from separate t -channel exchange (blue and red). Vertices are denoted by black dots. The diagrams at the top, which feature the hexagon of Fig. 6 as an externally labelled contraction minor have HRs, while the diagram at the bottom, which does not feature that hexagon as a contraction minor, does not.

The current certificates give

Graph	central-emission paths	NMRK with $w = z$	generic DSC
one-loop hexagon	same	1 full certificate	1 full certificate
planar hexbox	same	4 full + 3 boundary	1 full certificate
nonplanar hexbox	same	1 full + 2 boundary	no certificate
hexagon-pentagon	separate	no staged candidate	no staged candidate

These entries count certified vectors, not necessarily distinct physical regions. In the planar NMRK scan thirteen candidates reach the staged alignment analysis, but six fail the restored-layer certification; the seven entries in the table are those that survive the full native $\mathcal{F}$ audit. Within the tested NMRK set, however, the same-exchange graphs have an HR and the separate-exchange graph does not. This is evidence for a topology criterion, not yet a theorem; the DSC nonplanar-hexbox result also shows that the topology is not sufficient in every kinematic expansion.

The crossed ordering is the six-point Mandelstam region in which Regge-cut effects arise. Bartels’ reggeon calculus organized high-energy amplitudes by one- and two-reggeon states in successive t -channels [5]. Bartels, Lipatov and Sabio Vera later showed that, for six and more legs, the BDS ansatz is incompatible with the Regge cuts predicted by the BFKL approach in appropriate physical channels [6], and computed the leading-logarithmic color-octet cut contribution and the corresponding $2 \rightarrow 4$ and $3 \rightarrow 3$ discontinuities [6]. That correction is the high-energy contribution that must reside in the six-point BDS

remainder function, and it initiated the extensive subsequent study of remainder functions in multi-Regge kinematics.

It is therefore natural to conjecture that the HR found here is the parameter-space origin of the relevant t -channel cut. This connection is not established by the present certificates alone. It requires matching the local positive pinch and its region integral to the appropriate Mandelstam discontinuity and multi-reggeon exchange. Enlarging the topology sample in Fig. 7 is a direct way to test the conjecture.

3.3.3 NMRK with the additional surface $w = z$

Use the MSLCV chart

$$X_{34} = \frac{\hat{X}_{34}}{\eta}, \quad X_{56} = \frac{\hat{X}_{56}}{\eta}, \quad X_{45}, Q, z, \bar{z}, w, \bar{w} = O(1), \quad \eta \rightarrow 0^+, \quad (3.98)$$

with

$$Q, \hat{X}_{34}, X_{45}, \hat{X}_{56}, K_z > 0, \quad K_z = (1 - z)(1 - \bar{z}) > 0. \quad (3.99)$$

On this physical NMRK sheet $z, \bar{z}$ and $w, \bar{w}$ are conjugate pairs. The restriction $w = z, \bar{w} = \bar{z}$ is additional to NMRK. For generic NMRK the relevant bilinear determinant is proportional to $(w - z)(\bar{w} - \bar{z})$, so the HR is absent.

On the restricted surface define

$$L_1^N = (1 + X_{45})x_4 - \hat{X}_{34}X_{45}x_5, \quad L_2^N = (1 + X_{45})x_2 - K_z X_{45} \hat{X}_{56} x_1. \quad (3.100)$$

The selected face has

$$\mathcal{F}_*^N = \mathcal{F}_{\text{SL}}^N + \mathcal{F}_{\text{Obs}}^N = \textcolor{red}{Q} \hat{X}_{34} \hat{X}_{56} \textcolor{red}{L}_1^N \textcolor{red}{L}_2^N + 0. \quad (3.101)$$

The two-stage vectors and certified answer are

$$\begin{aligned} \phi_N &= (-1, -1, -2, -1, -2, -1), & \rho_N &= (-1, -1, -1, -1, -1, -1), \\ \vec{v}_{\text{HR}}^N &= (-2, -2, -3, -2, -3, -2; 1), & (W_{\text{SL}}, W_{\text{HR}}) &= (-4, -3). \end{aligned} \quad (3.102)$$

There is no intermediate cancellation layer. At W_{HR} ,

$$\begin{aligned} \mathcal{U}_{-3} &= \textcolor{blue}{x}_2 + \textcolor{blue}{x}_4, \\ \mathcal{F}_{-3} &= -\textcolor{blue}{Q} \hat{X}_{34} X_{45} \hat{X}_{56} (\textcolor{blue}{K}_z x_0 x_2 + \textcolor{blue}{K}_z x_0 x_4 + \hat{X}_{34} x_3 x_5 + \textcolor{blue}{K}_z \hat{X}_{56} x_1 x_3). \end{aligned} \quad (3.103)$$

These monomials give a unique lower facet directly. The required new capability is asymptotic-order alignment itself; once the correct face is exposed, the cancellation has depth one. Restoring and testing every native η -layer leaves this one-loop certificate unchanged.

3.3.4 Generic double spacelike-collinear kinematics

For DSC let $\varepsilon \rightarrow 0^+$ and use the real physical slice

$$a < 0, \quad -1 < \tau_1 < 0, \quad \tau_2 < -1, \quad z_D < 0, \quad \bar{z}_D < 0. \quad (3.104)$$

The symbols $z_D, \bar{z}_D$ are conjugate on the standard real transverse MRK branch, but after continuation to this DSC sheet they are two independent real roots. In the graph-polynomial sign convention the adjacent invariants behave as

$$\begin{aligned} s_{12} &= -\frac{\tau_2}{a(1+\tau_1)(1+\tau_2)} + O(\varepsilon), & s_{23} &= \frac{\varepsilon^2 \tau_1}{a \bar{z}_D} + O(\varepsilon^3), \\ s_{34} &= \frac{1}{(a-1)(1+\tau_2)} + O(\varepsilon), & s_{45} &= -\frac{1}{a}, \\ s_{56} &= \frac{\tau_1}{(a-1)(1+\tau_1)} + O(\varepsilon), & s_{16} &= \frac{\varepsilon^2 \tau_2 z_D}{a} + O(\varepsilon^3). \end{aligned} \quad (3.105)$$

Thus $s_{23}, s_{16} < 0$, while $s_{12}, s_{34}, s_{45}, s_{56} > 0$, as required for $2 \rightarrow 4$ scattering with legs 1 and 2 incoming.

The cancellation factors and selected-face decomposition are

$$L_1^D = (1+\tau_2)x_4 + x_5, \quad L_2^D = \tau_1 x_1 + (1+\tau_1)x_2, \quad (3.106)$$

$$\mathcal{F}_\star^D = \mathcal{F}_{\text{SL}}^D + \mathcal{F}_{\text{Obs}}^D = -(a-1)a^2 \bar{z}_D^2 L_1^D L_2^D + 0. \quad (3.107)$$

Both factors have positive zeros,

$$x_5 = -(1+\tau_2)x_4 > 0, \quad x_2 = -\frac{\tau_1}{1+\tau_1}x_1 > 0. \quad (3.108)$$

For crossed double-timelike collinear splittings $\tau_i > 0$, so these same factors have one sign for positive x_i and the pinch disappears.

The alignment vector and the provisional face-only HRF vector are

$$\phi_D = (-2, -2, -2, -1, -2, -2), \quad \rho_D^{(0)} = (-1, -1, -1, -1, -1, -1). \quad (3.109)$$

The provisional vector is obtained from the isolated aligned face; it is not the second vector in the final composition. With $\mathcal{I}_D = \langle L_1^D, L_2^D \rangle$, the actual occupied layers obey

$$\underbrace{\mathcal{F}_{-4} \in \mathcal{I}_D^2}_{\text{ord}_{\mathcal{I}_D}=2} + \underbrace{\mathcal{F}_{-3} \in \mathcal{I}_D \setminus \mathcal{I}_D^2}_{\text{ord}_{\mathcal{I}_D}=1} + \underbrace{\mathcal{F}_{-2} \notin \mathcal{I}_D}_{\text{ord}_{\mathcal{I}_D}=0}. \quad (3.110)$$

The middle layer contains 52 nonzero terms. It is not a missing power of ε : it vanishes to first order on the common pinch. The leading $\mathcal{U}$ layer is

$$\mathcal{U}_{-2} = x_0 + x_1 + x_2 + x_4 + x_5, \quad (3.111)$$

and the full ideal-jet balance gives

$$\begin{aligned} \rho_D &= (0, 0, 0, 1, 0, 0), & \phi_D + \rho_D &= (-2, -2, -2, 0, -2, -2), \\ \vec{v}_{\text{HR}}^D &= (-2, -2, -2, 0, -2, -2; 1), & (W_{\text{SL}}, W_{\text{HR}}) &= (-4, -2), \end{aligned} \quad (3.112)$$

with gap two. A local dissection pulls back to this same vector.

This example is why asymptotic-order alignment cannot be treated as a cosmetic preprocessing step. HRF must (i) expose the cancelling face, (ii) retain every occupied layer, (iii) compute successive ideal orders, and (iv) use the first nonvanishing LP layer to fix the uniform ambiguity of the face representative. Simply adding $\phi_D + \rho_D^{(0)}$ would give the wrong total vector and the wrong cancellation depth. This is the concrete example for the distinction between $\rho^{(0)}$ and the corrected ρ in Sec. 2.3.

3.3.5 What the two limits share—and what they do not

After their separate kinematic rescalings, the factor coefficients are identified by

$$\frac{K_z X_{45} \hat{X}_{56}}{1 + X_{45}} = -\frac{\tau_1}{1 + \tau_1}, \quad \frac{\hat{X}_{34} X_{45}}{1 + X_{45}} = -\frac{1}{1 + \tau_2}. \quad (3.113)$$

Hence the two order-aligned $\mathcal{F}_{\text{SL}}$ supports describe the same type of positive cancellation hypersurface. The next layers differ:

Expansion	certified $\mathbf{v}$	$(W_{\text{SL}}, W_{\text{HR}})$	depth
NMRK with $w = z$	$(-2, -2, -3, -2, -3, -2)$	$(-4, -3)$	1
generic DSC	$(-2, -2, -2, 0, -2, -2)$	$(-4, -2)$	2

The common generator does not make the two limits identical. The first resolved LP layer, and therefore the physical region vector, remembers the path by which the singular hypersurface is approached.

What the examples teach

Family	New structure	Capability needed not to miss the HR
Crown	simultaneous positive cancellation, $\mathcal{F}_{\text{Obs}} = 0$	keep a multi-equation positive pinch and determine its uniform scaling
Crown descendants	contraction-minor boundary regions	scan boundary strata and compare restricted ideals, not only parent interiors
Five-point seed	$\mathcal{F}_{\text{Obs}} \neq 0$, non-uniform scaling	isolate $\mathcal{F}_{\text{SL}}$ before imposing the positive pinch; retain the obstruction
Vertex correction	primitive non-binomial factor	generate and test general polynomial ideals rather than monomial pairs only
Five-point near-planar wide angle	three path normals and a three-generator pair-product ideal	allow more than two polynomial generators and certify the pullback from local coordinates
Five-point central-soft MRK	a two-factor cancellation exposed on a nontrivial alignment face	preserve every native kinematic power when composing the vectors; certify the resolved full-LP support
NMRK, $w = z$	cancellation on a nontrivial asymptotic face	align orders before running the ordinary HRF construction
DSC	a second occupied cancellation layer	resolve the ideal filtration and fix the uniform face ambiguity from the full LP polynomial

The progression is cumulative. Later examples require the earlier tests in addition to their new feature. In every case the final statement is again geometric: after cancellation is resolved, the leading polynomial must be a lower facet in coordinates adapted to the pinch. HRF reconstructs the coefficient and Landau information absent from the original Newton polytope; it does not replace lower-facet certification.

Allowing massive internal propagators would relax the standing massless multiaffinity assumption. The resulting quadratic edge-parameter dependence changes the possible

cancellation jets and the local facet problem. It is a genuine extension of this framework, not another example in the present tour.

A Notation summary

Symbol	Meaning
δ	kinematic expansion parameter
$\mathbf{s}$	real kinematic coordinates; their physical domain $\mathcal{K}$ is specified by explicit inequalities
x_e	independent LP edge parameter, integrated over $(0, \infty)$
$\mathcal{C}_{\text{LP}} = \mathbb{R}_{>0}^N$	non-projective LP integration contour
$\mathcal{P} = \mathcal{U} + \mathcal{F}$	Lee–Pomeransky polynomial
$\deg_x \mathcal{U} = L, \deg_x \mathcal{F} = L + 1$	massless graph-polynomial degrees; both polynomials are linear in each individual x_e
$\widehat{\mathcal{P}} = \mathcal{F} + x_0 \mathcal{U}$	optional homogeneous $(N + 1)$ -parameter embedding
$m_i = c_i(\mathbf{s}) \delta^{a_i} \mathbf{x}^{\mathbf{r}_i}$	monomial of $\mathcal{P}$
$\mathbf{r}_i$	length- N LP-parameter exponent vector of m_i
$\vec{r}_i = (\mathbf{r}_i; a_i)$	augmented exponent point
$\mathcal{F}_0$	strict fixed- $\mathbf{x}$ leading polynomial
$\mathcal{F}^{[\phi]}$	face selected by the preliminary alignment vector ϕ
$x_e = \delta^{\phi_e} y_e$	first transformation: asymptotic-order alignment
$\mathcal{F}_\star$	HRF starting polynomial, either $\mathcal{F}_0$ or $\mathcal{F}^{[\phi]}$
$\mathcal{S}_{\text{SL}}$	superleading monomial support selected by the scaling law
$\mathcal{F}_{\text{SL}}$	superleading cancellation sector
$\mathcal{F}_{\text{Obs}}$	obstruction polynomial
f_j	non-monomial cancellation factor
$\mathcal{I}$	ideal of independent cancellation factors at the common pinch
$\rho^{(0)}$	provisional face-only HRF scaling in the aligned variables
$y_e = \delta^{\rho_e} z_e$	corrected second-stage HRF scaling after all LP layers are restored
$\mathbf{v}_{\text{HR}} = \phi + \rho$	composition of the two edge-parameter transformations
$\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1)$	certified total hidden-region vector; the last component is appended once and is not added stage by stage
W_{SL}	common individual weight of SL monomials
W_{HR}	first resolved nonvanishing LP weight after cancellation
$W_{\text{HR}} - W_{\text{SL}}$	cancellation depth
red terms	individually superleading sector at W_{SL}
blue terms	resolved physical leading layer at W_{HR}
black terms	all other layers, including intermediate ideal-valued cancellations

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